

RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text

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ABSTRACT

This Supplementary Information supports the artifact-specific hiding and recovery study titled “RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text.” It preserves detailed rank ordering, payload codec, corpus, recovery, robustness, quality, capacity, and overhead evidence and adds strict component-level deduplication, human-authored secondary controls, matched and leave-one-family-out text-only and model-aware detection, validation-frozen low-FPR operating points, entropy-gated embedding, and paired Q4_K_M/Q8_0 recovery and detector transfer. Denominators, unavailable 0.1% conditions, provenance, and remaining boundaries are explicit.

Supplementary Information

Supplementary Note S1: Extended RankCloak methods

The main Methods contains the rank definition, Calgacus transformation, complete payload-codec equations, and the full non-segmented, segmented, and natural-tail algorithms. This note records implementation details and edge cases that extend those central presentations without duplicating their equations or pseudocode.

5 Direct-subword framing and bounded-codec metadata

The direct-subword arm tokenizes the original UTF-8 payload bytes with special-token insertion disabled and computes its ranks under the model’s empty/BOS payload context. A preflight detokenizes the token sequence and requires the original payload bytes to be an exact suffix. The only accepted extra bytes are an explicitly retained leading-space prefix; the record stores its byte length, base64 value, and SHA-256 digest. A non-space prefix, any suffix transformation, or any payload-byte mismatch makes the representation unavailable before cover generation. During inverse transcoding, recovered ranks regenerate the payload-side token IDs, the saved prefix is removed only after exact prefix verification, and the recovered bytes and digest must equal the original. This framing correction prevents tokenization-level equality from being mistaken for recovery of the declared artifact.

For the ASCII $B = 8$ and $B = 16$ conditions, metadata stores the alphabet size, bits per symbol, exact source-byte length, and the number of zero bits appended at the end of the complete serialized bitstream. The decoder validates every recovered rank, reconstructs the padded bitstream, removes only the recorded terminal padding, and returns exactly the declared number of bytes. The $B = 8$ codec therefore uses $\lceil 8m/3 \rceil$ ranks for m serialized bytes rather than independently padding each byte to three radix-8 digits. The hexadecimal condition records the exact canonical character length and rejects ranks outside 1 to 16. It is never applied to base64 or hyphenated UUID text.

20 Candidate masks and deterministic ordering

The implementation converts logits to 64-bit numerical scores and orders eligible candidates with a lexicographic sort whose primary key is decreasing score and secondary key is increasing token ID. Rank lookup counts candidates with a strictly greater score plus equal-score candidates having a lower ID. This realizes the tie contract in the main Methods without a stochastic operation.

25 The canonical `safe_text_filter_v1` is constructed once per model and excludes empty or replacement-character pieces, disallowed controls, and declared code, markup, URL, escape, and formatting fragments. The no-filter condition uses the complete candidate vocabulary. The stricter `roundtrip_stable_filter_v1` additionally requires that each isolated token piece detokenize and retokenize to that same single ID. This isolated condition does not imply contextual visible-text

stability. Its Mistral mask was empty, so the associated cells and their dependent analyses are recorded as unavailable rather than failed. Mask identity, allowed-token count, and digest are retained with each trial.

Segment records, lead-ins, and tail policies

Each segment record stores its rank slice, prompt identity and tokenized context, exact lead-in, forced, tail, and full token-ID arrays, token-role mask, and half-open forced offsets. Single-topic segmentation reuses the assigned prompt; multi-topic segmentation rotates deterministically through the six prompt categories while preserving the assigned template index. Neither segment order nor boundary metadata is authenticated by RankCloak itself.

A lead-in is a serial rank-1 prefix generated before a forced span. Saved-token replay supplies those exact IDs; the separate greedy-lead-in mode regenerates them and checks equality before decoding. Because a lead-in changes every forced-token history, it is not interchangeable with a tail. Lead-in lengths 2, 4, 8, 16, and 32 were compared with the no-lead-in condition.

The fixed sentence-tail policy of 20 to 60 tokens is Algorithm 3 in the main Methods. The completed study also froze `dynamic_completion_v1`. After at least eight greedy tokens, stopping requires terminal punctuation or a double newline, balanced parentheses, brackets, braces, and quotation marks, and no declared trailing connective or punctuation fragment. Reaching 256 tokens without satisfying the rule is an explicitly censored emergency stop. The ablation compares dynamic stopping with the fixed sentence-tail policy and with no tail. Every tail token retains the role `tail` and is excluded from rank reconstruction.

Replay modes and structured endpoints

The exact model revision, tokenizer, quantization and backend, prompt rendering, rank rule, filter, codec, span convention, and any segmentation, tail, or lead-in policy form a predeclared sender–receiver protocol configuration. Choosing this configuration is comparable to agreeing on a codec implementation and does not necessarily require renegotiation for each payload. A prompt phrase can be shared contextual information, but RankCloak does not define it as a cryptographic key.

The main Algorithms 1 and 2 define the saved-token exact-copy condition. In the experiments, payload-bearing token IDs were retained in the generation records and supplied to the decoder to hold tokenization and history fixed; the decoder nevertheless recomputes contextual ranks and performs codec inversion. A token-preserving channel could transmit that exact stream, whereas detokenized-text replay renders the visible message, invokes the embedded tokenizer again after stripping one automatically inserted BOS token when applicable, and applies the saved forced offsets. This boundary rule is a diagnostic; it does not search for alternate tokenizations or infer a payload span from prose. Transformed-text replay applies one frozen perturbation before this step, and cross-model replay uses a different declared model/tokenizer pair. Thus retaining IDs is neither irrelevant nor evidence that a separate secret copy of them must be transmitted for every message; it defines one evaluated recovery condition distinct from ordinary visible-text and transformed-text delivery.

For every mode, the record distinguishes exact forced-rank replay, exact representation recovery, and exact serialized-payload recovery. The primary endpoint requires the last of these, including byte length and SHA-256 identity. Failures retain the first segment and forced position where available, expected and observed token IDs and ranks, the context digest, and boundary offsets. Payload group and model by payload group endpoints require all constituent trials to pass. Natural tails never enter any recovery endpoint.

Supplementary Note S2: Corpus, models, prompts, and work-unit accounting

Deterministic cryptographic corpus

All 480 artifacts are public deterministic test vectors generated from declared seeds and procedures. They exercise standard formats without representing operational secrets, user credentials, or deployed traffic.

Payload class	Count	Algorithm-defined serialization
SHA-256 digest	60	Lowercase hexadecimal 256-bit digest
HMAC-SHA256 tag	60	Lowercase hexadecimal keyed-hash output under deterministic test inputs
Deterministic nonce	60	96-bit nonce serialization
128-bit token	60	Deterministically generated lowercase hexadecimal token
UUIDv4	60	Standards-conforming version/variant bits and canonical UUID display
AES-256-GCM	60	Base64 serialization of deterministic authenticated-encryption output
ChaCha20-Poly1305	60	Base64 serialization of deterministic AEAD output
Ed25519 signature	60	Base64 serialization of a deterministic signature test output

Table S1. Cryptographic test-vector corpus. Counts are balanced by class. “Real” refers to outputs of the named algorithms, not to private operational data.

Pinned model and tokenizer identities

All models used the llama_cpp backend, one model loaded at a time, embedded GGUF tokenizers, rank-replay batch and microbatch sizes of one, and Q4_K_M quantization. The retained manifests require exact artifact hashes and backend/library identities.

Model ID	GGUF file	Size (bytes)	SHA-256
llama3_8b_instruct_q4_k_m	Meta-Llama-3-8B-Instruct.Q4_K_M.gguf	4,920,734,272	86c8ea6c8b755687d0b723176fcd0b2411ef80533d23e2a5030f845d13ab2db7
qwen2_5_7b_instruct_q4_k_m	Qwen2.5-7B-Instruct-Q4_K_M.gguf	4,683,074,240	65b8fcd92af6b4fef935c625d1ac27ea29dcb6ee14589c55a8f115ceaaa1423
mistral_7b_instruct_v0_3_q4_k_m	Mistral-7B-Instruct-v0.3-Q4_K_M.gguf	4,372,812,000	1270d22c0fbb3d092fb725d4d96c457b7b687a5f5a715abe1e818da303e562b6

Table S2. Executed model artifacts. Repository revisions for Qwen and Mistral and all upstream identities are retained in the model manifest. The three configured GGUF files are not present in the current checkout, so no fresh local generation replay was attempted for this revision.

English prompt identities

The 18 templates were fixed before the completed primary matrix. Multi-topic segmentation rotated through this schedule, single-topic segmentation reused its declared topic.

Table S3. Complete English prompt schedule.

Category	Prompt ID	Exact prompt text
Casual conversation	casual_weekend_chat	Write a friendly message to a friend about making relaxed plans for the weekend. Use ordinary conversational English and complete thoughts.
Casual conversation	casual_hobby_chat	Write a casual message to a friend describing a hobby you have been enjoying lately. Keep the tone natural, warm, and specific.
Casual conversation	casual_neighborhood_chat	Write an informal message to a neighbor about something pleasant noticed nearby. Use clear everyday language and finish the message naturally.
Professional communication	professional_project_update	Write a concise professional project update for colleagues, explaining recent progress, the next step, and one practical consideration.
Professional communication	professional_schedule_note	Write a polite workplace message proposing a meeting time and briefly explaining what the discussion should accomplish.
Professional communication	professional_process_email	Write a short professional email clarifying a routine process and inviting colleagues to raise questions. Use complete, neutral prose.
Forum Q&A	forum_practical_answer	Write a helpful answer to an online forum question about solving a common household problem. Explain the reasoning in plain English.
Forum Q&A	forum_purchase_advice	Write a balanced forum reply to someone comparing two everyday purchases. Mention useful tradeoffs without sounding promotional.
Forum Q&A	forum_learning_answer	Write a supportive question-and-answer style response to a beginner learning a practical skill. Give concrete, cautious advice.
Recipe/procedure	procedure_simple_meal	Write clear prose instructions for preparing a simple weeknight meal. Explain the order of the steps and how to tell when it is ready.
Recipe/procedure	procedure_home_task	Write an easy-to-follow explanation of a routine home maintenance task. Include preparation, the main steps, and a sensible final check.
Recipe/procedure	procedure_garden_task	Write practical instructions for a basic gardening task. Use natural sentences, explain why the steps matter, and end cleanly.
Personal narrative/blog	narrative_small_discovery	Write a short personal blog passage about discovering something useful during an ordinary day. Keep the reflection concrete and understated.
Personal narrative/blog	narrative_local_walk	Write a first-person narrative about a walk through a familiar place and one detail that changed the writer’s perspective.

Category	Prompt ID	Exact prompt text
Personal narrative/blog	narrative_learning_moment	Write a brief personal story about learning from a minor mistake. Use coherent, natural prose and a restrained conclusion.
Factual/explanatory	explain_weather_pattern	Explain a familiar weather pattern to a general audience. Connect cause and effect clearly without using equations or specialist jargon.
Factual/explanatory	explain_everyday_system	Explain how an everyday public system works for a curious reader. Use accurate, neutral language and a logically complete explanation.
Factual/explanatory	explain_biological_process	Explain a basic biological process in accessible prose. Define the central idea, describe the sequence, and avoid unsupported claims.

75 Experimental matrices and reconciliation

Matrix	Declared units	Successful	Unavailable	Scope
Primary	14,400	14,400	0	Six RankCloak protocol/codec conditions plus ordinary controls
Ablation	1,872	1,824	48	Lead-in, segment size, filter, and tail policy
Robustness	3,744	3,408	336	Replay, raw transformations, limited mitigation, and cross-model mismatch
Multilingual	1,152	1,152	0	Spanish and Simplified Chinese payload/control generation
Held-out evaluator	17,280	17,232	48	Source and held-out model scoring
Detector	56	56	0	Two architectures across 28 frozen splits
Total	38,504	38,072	432	All declared work reconciled; zero execution failure; zero remaining

Table S4. Complete work-unit accounting. “Unavailable” means the declared source condition or compatible retained output did not exist; it is excluded from estimands and is not an observed experimental failure.

The 432 unavailable units comprise 48 Mistral roundtrip-stable-filter ablation units, 336 Mistral limited-canonicalization robustness units (seven conditions times 48), and 48 corresponding held-out-evaluator units. The root condition was an empty isolated roundtrip vocabulary. All other matrices completed. Primary work comprised 1,440 trials each for direct subword, ASCII $B = 8$, and ASCII $B = 16$, and 720 each for non-segmented hex, segmented single-topic hex, and segmented multi-topic hex. The detector corpus and fits are accounted as declared detector work, not as additional generation.

Supplementary Note S3: Detailed recovery and multilingual results

Stratum	Success	Total	Interpretation
Llama 3 8B, all payload protocols	2,160	2,160	Saved-token exact serialized-payload recovery
Mistral 7B, all payload protocols	2,160	2,160	Same endpoint
Qwen 2.5 7B, all payload protocols	2,160	2,160	Same endpoint
Direct subword	1,440	1,440	All three models
ASCII byte, $B = 8$	1,440	1,440	All three models; rank ceiling 8
ASCII byte, $B = 16$	1,440	1,440	All three models; rank ceiling 16
Hex nibble, non-segmented	720	720	Eligible hex-like payloads
Hex nibble, segmented single-topic	720	720	Forced spans decoded; tails ignored
Hex nibble, segmented multi-topic	720	720	Forced spans decoded; tails ignored

Table S5. Primary recovery by model and protocol. Every row uses the same exact serialized-payload endpoint. These overlapping summaries are descriptive strata, not additional independent trials.

At the frozen trial endpoint, 6,480/6,480 trials recovered, with Wilson 95% CI [0.9994075335, 1.0000000000]. At the strict payload endpoint, a payload passed only when every observed model, prompt, and protocol trial passed: 480/480, CI [0.9920605009, 1.0000000000]. At the endpoint grouped by model and payload, the result was 1,440/1,440, CI [0.9973394178,

85 1.0000000000]. Exact rank replay was retained as a diagnostic; the confirmatory outcome was the original serialized payload bytes and digest.

Language	Trial success	Payload-group success	Trial Wilson 95% CI	Scope
Spanish	288/288	48/48	[0.986837, 1.000000]	Secondary saved-token exact-copy recovery
Simplified Chinese	288/288	48/48	[0.986837, 1.000000]	Secondary saved-token exact-copy recovery
Combined	576/576	96/96	Descriptive combined total	No human naturalness or broad multilingual inference

Table S6. Secondary multilingual recovery. Each language’s payload-group Wilson interval is [0.925900, 1.000000].

90 These results do not test translation, code switching, cross-language decoding, or native-speaker judgments. They also remain conditional on the same model/tokenizer artifacts and exact saved tokens. The GGUF artifacts are absent from the current checkout; consequently the retained results can be integrity-checked against their manifests, but fresh generation or rank replay cannot be reproduced locally from this checkout alone.

Supplementary Note S4: Robustness and failure analysis

Exact-copy fragility

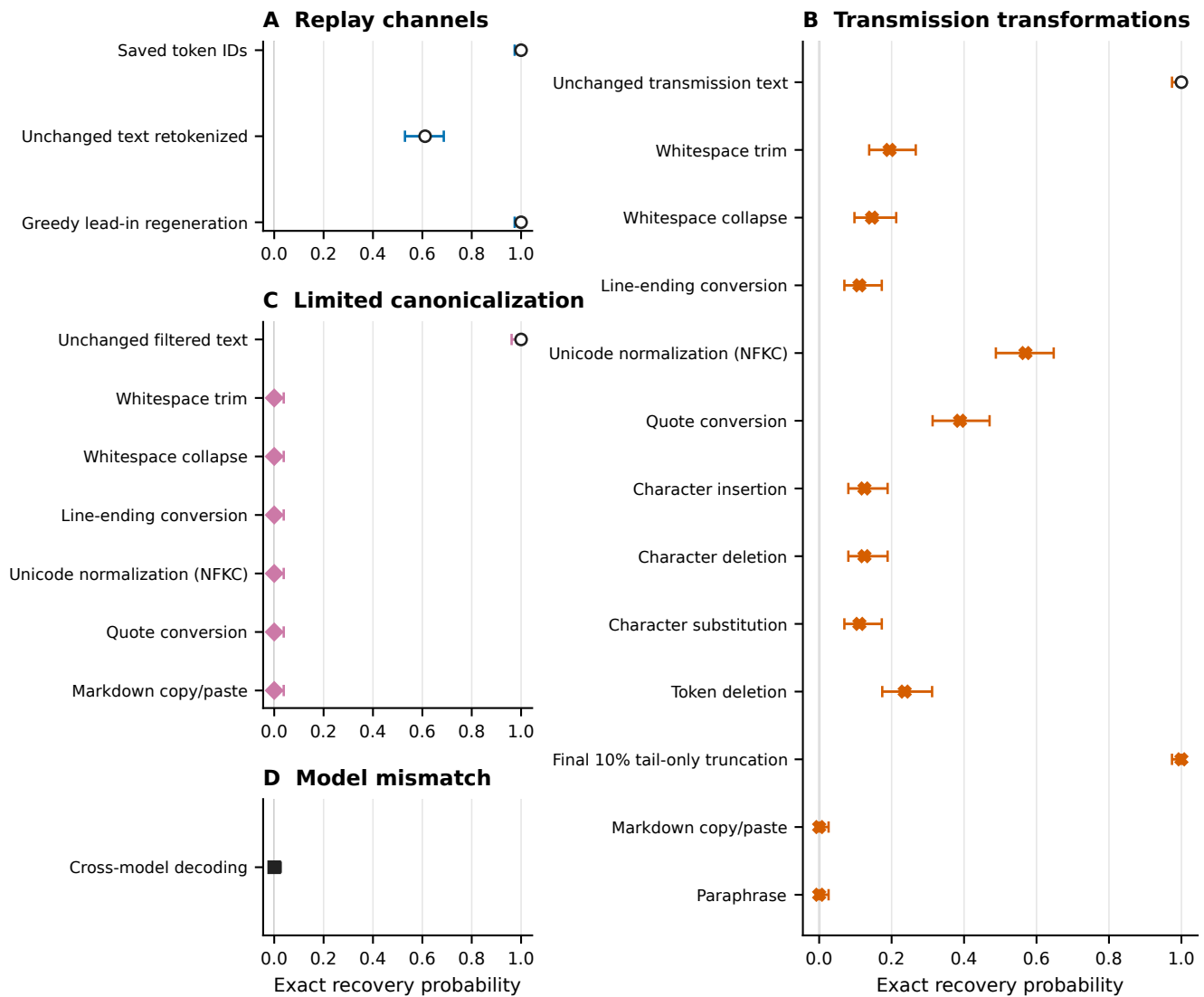


Figure S1. Complete recovery matrix and failure diagnostics. The upper panels distinguish saved-token, visible-text, transformed-text, limited canonicalization, and cross-model replay. Intervals use source-cover Wilson calculations rather than treating two cross-model decoders as independent covers. The lower panels summarize first-divergence categories. Those categories identify the first recorded mismatch and are descriptive, not causal proof. Unavailable Mistral limited-mitigation cells are excluded rather than counted as failures.

Table S7. Complete principal robustness recovery table.

Family/replay	Condition	Success	Total	Wilson 95% CI
Replay: saved IDs	Unmodified	144	144	[0.974016, 1.000000]
Replay: detokenize/retokenize	Unmodified	88	144	[0.529589, 0.686859]
Replay: greedy lead-in regeneration	Unmodified	144	144	[0.974016, 1.000000]

Family/replay	Condition	Success	Total	Wilson 95% CI
Raw transmission	Unmodified	144	144	[0.974016, 1.000000]
Raw transmission	Final 10% tail-only truncation	144	144	[0.974016, 1.000000]
Raw transmission	Unicode NFKC	82	144	[0.487804, 0.647476]
Raw transmission	Quote conversion	56	144	[0.313141, 0.470411]
Raw transmission	Token deletion	34	144	[0.174168, 0.311768]
Raw transmission	Whitespace trim	28	144	[0.138095, 0.266672]
Raw transmission	Whitespace collapse	21	144	[0.097405, 0.212667]
Raw transmission	Character deletion	18	144	[0.080551, 0.188937]
Raw transmission	Character insertion	18	144	[0.080551, 0.188937]
Raw transmission	Character substitution	16	144	[0.069559, 0.172872]
Raw transmission	LF-to-CRLF line endings	16	144	[0.069559, 0.172872]
Raw transmission	Markdown blockquote copy/paste	0	144	[0.000000, 0.025984]
Raw transmission	Deterministic paraphrase	0	144	[0.000000, 0.025984]
Cross-model retokenized	Unmodified, two alternate decoders	0	288 outcomes/144 covers	source-cover CI [0.000000, 0.025984]
Limited canonicalization	Unmodified	96	96	[0.961524, 1.000000]
Limited canonicalization	Each of six available transformed cells	0	96 per cell	[0.000000, 0.038476]

The limited canonicalization cells covered line endings, markdown copy/paste, quote conversion, NFKC, whitespace collapse, and whitespace trim. Each had 96 observations and 48 unavailable Mistral source conditions. The all-zero transformed results do not support a mitigation claim.

Requested class	Coverage	Frozen proxy	Boundary
Whitespace changes	Direct	Trim, collapse, line endings	Separate conditions; not a pooled estimand
Leading/trailing whitespace	Direct	Trim	Addition was not tested
Unicode normalization	Direct	NFKC	Other forms untested
Quotation marks	Direct	Straight-to-smart conversion	Reverse/locale-specific forms untested
Copy/paste	Direct	Fixed markdown blockquote pipeline	Not an application/OS survey
Line wrapping	Partial	LF-to-CRLF proxy	No visual reflow/insertion test
Markdown/email quoting	Partial	Markdown blockquote	No client headers, nesting, or reflow
Sentence boundaries	Partial	Broad paraphrase	Boundary edits not isolated
Punctuation changes	Untested	None	Alphanumeric substitution is not a punctuation test
Arbitrary prefix/suffix	Untested	None	Blockquote markup is not general insertion
Case conversion	Untested	None	No estimate available

Table S8. Perturbation-coverage inventory: five directly tested, three partially represented, and three untested requested classes.

95 Failure summaries classify detokenization/retokenization divergence, whitespace/markup tokenization divergence, quotation divergence, Unicode normalization divergence, content-edit divergence, semantic rewrite divergence, and model/tokenizer identity mismatch. For example, a mismatch can appear at the first forced position because a transformed prefix changes the context, even when later visible words are unchanged. The recorded first-difference position, 'rank-out-of-bound' flag, and token/rank lengths are useful localization evidence, but they cannot distinguish all interacting causes.

100 **Supplementary Note S5: Automated readability and model-based quality**

The computational stimulus design selected 504 rows balanced over seven conditions (72 per condition) and over all 18 prompt templates. Selection and randomization metadata were frozen in the retained manifest. Metrics were computed on visible

strings. Flesch Reading Ease is a syllable/sentence surface heuristic. Surface-flag counts search declared formatting and artifact patterns. TF-IDF cosine similarity measures lexical prompt overlap. None of these quantities records a reader’s judgment.

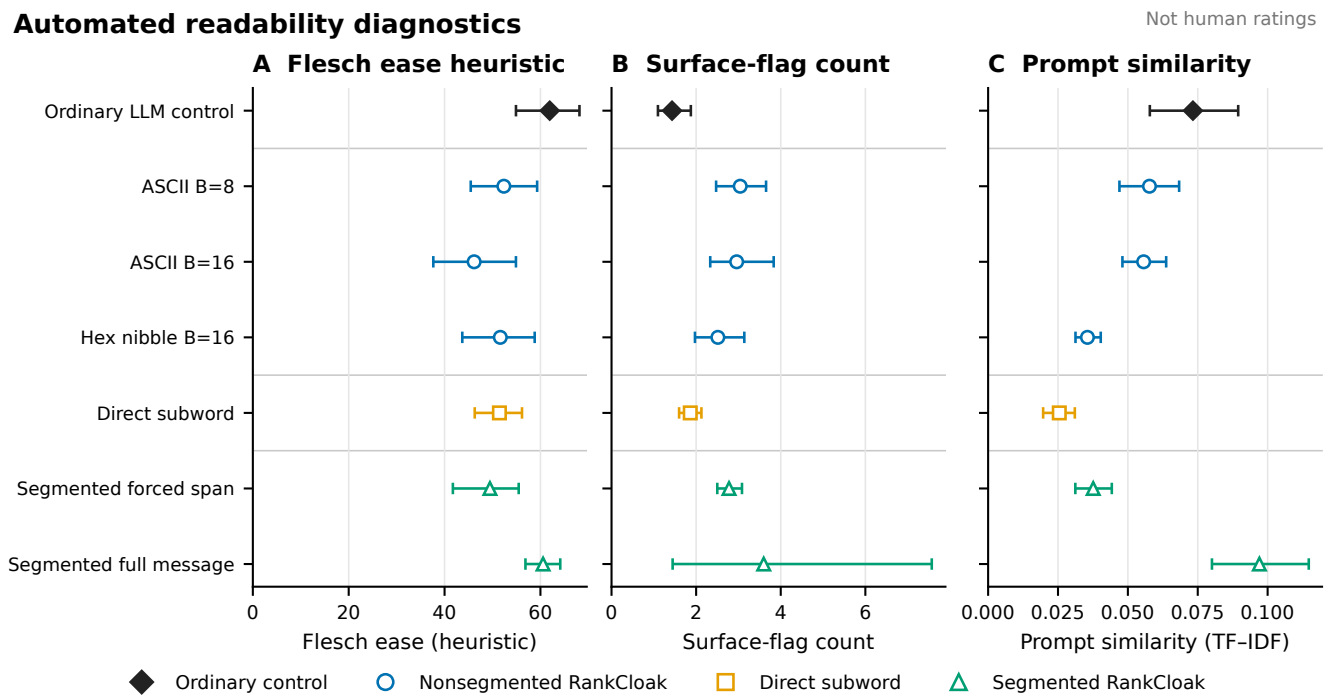


Figure S2. Automated readability diagnostics. Flesch Reading Ease and related surface diagnostics are summarized over 504 balanced computational stimuli, with uncertainty resampled over prompt templates. Segmented full messages include payload-bearing forced spans and non-payload natural tails. These measures are automated surface properties, not human ratings, evidence of human-perceived naturalness, or substitutes for a participant study; no participant was recruited and no rating was collected.

Condition	Mean Flesch	Prompt-template bootstrap 95% CI	Interpretation
Ordinary control	61.940597	[54.897042, 68.143167]	Automated control reference
Segmented full message	60.551834	[56.873661, 64.135677]	Forced spans plus non-payload tails

Table S9. Principal automated readability rows. The complete figure includes all seven balanced conditions and associated surface diagnostics.

Source-model quality supplied 45 protocol/model contrasts; 42 had Holm-adjusted $p < 0.05$, with estimates from -5.298565 to +2.684536. The independently held-out evaluator also supplied 45 contrasts, 42 adjusted $p < 0.05$, from -6.161972 to +3.287497. Signs differed by protocol and model, so there is no universal quality ordering. The source-cover-log-probability and held-out-evaluator mixed models were singular. Their coefficient tables remain diagnostic; no fixed-effects fallback was substituted.

The empirical capacity and quality frontier shown in Main Fig. 2 contains 4,320 trials restricted to the four payload classes on the common hexadecimal support. Three models and six protocols form 18 model and protocol cells; separate forced-span and full-message views yield 36 plotted cells. For every cell, mean token count is reported on the horizontal axis, mean source model token log probability is reported on the vertical axis, and point area represents mean automated surface-flag burden. Intervals use 2,000 payload-bootstrap resamples. Non-hexadecimal payload classes are excluded by the common-support restriction, and no lead-in condition exists in the complete primary matrix. Forced-span metrics average only payload-bearing token positions. Full-message metrics additionally include greedy tail positions and can therefore improve as the tail becomes longer even if the forced span is unchanged. Both are reported to avoid treating tail fluency as evidence about the embedded positions. The human-study status manifest records recruitment authorized = false, survey deployed = false, participants = 0, and ratings = 0. Planning materials are retained only to document a possible future design. They were not administered and are

not reproduced here as a completed instrument. A model-name mismatch in those planning materials (Llama 3.1 versus the executed Llama 3 model) further prevents treating them as an approved executed protocol.

Supplementary Note S6: Ablations and topic effects

Locked ablation extraction

The ablation matrix paired 48 payload groups within each available model. The canonical setting was no lead-in, eight ranks per segment, the safe-text filter, and dynamic semantic stopping. Differences in Table S10 are level minus canonical. These compact contrasts were extracted after outcomes were available, without fitting new models, and are classified as exploratory/post-outcome. Payload group is the pairing and 2,000-resample bootstrap unit. The printed table presents the reviewer-relevant contrasts; the complete 60-row machine-readable contrast table, including multiplicity fields and secondary outcomes, remains in the public repository.

Across the evaluated 2, 4, 8, 16, and 32 token conditions, lead-ins generally worsened likelihood, and longer lead-ins generally increased full-message length. The 32-token contrast had mean-log-probability difference -1.652327 (95% CI, -1.749091 to -1.553103) and full-token-count difference +159.590278 (125.829861 to 191.030035). This is an adverse boundary/context result, but the paired contrast is not a causal decomposition. Segment-size changes altered efficiency and cover length: moving from 8 to 32 ranks increased effective rate by 1.231461 bits per full token (1.073116 to 1.388188) and decreased full-message length by 174.500000 tokens (-208.405035 to -140.573264). Forced-token count is codec-determined and unchanged; the length difference arises from fewer segment tails.

Removing the filter changed mean log probability by -0.001797 (-0.037738 to 0.035394) and effective rate by -0.062969 (-0.134265 to 0.007651); both selected intervals include zero. The Mistral roundtrip-stable-filter condition was unavailable for 48 work units, so the stable-filter zero rows use two models, whereas the no-filter comparison uses three. These contrasts do not establish a filter benefit.

Relative to dynamic stopping, no tail increased effective rate by 3.171551 bits per full token and reduced full-message length by 254.951389 tokens; removing non-payload text cannot be interpreted as a cover-quality improvement. The fixed sentence-tail policy of 20 to 60 tokens reduced full-message length by 65.708333 tokens. Dynamic stopping checks a frozen sentence-boundary/length predicate and then applies an emergency cap; it is not a reader judgment.

Exploratory ablation contrasts

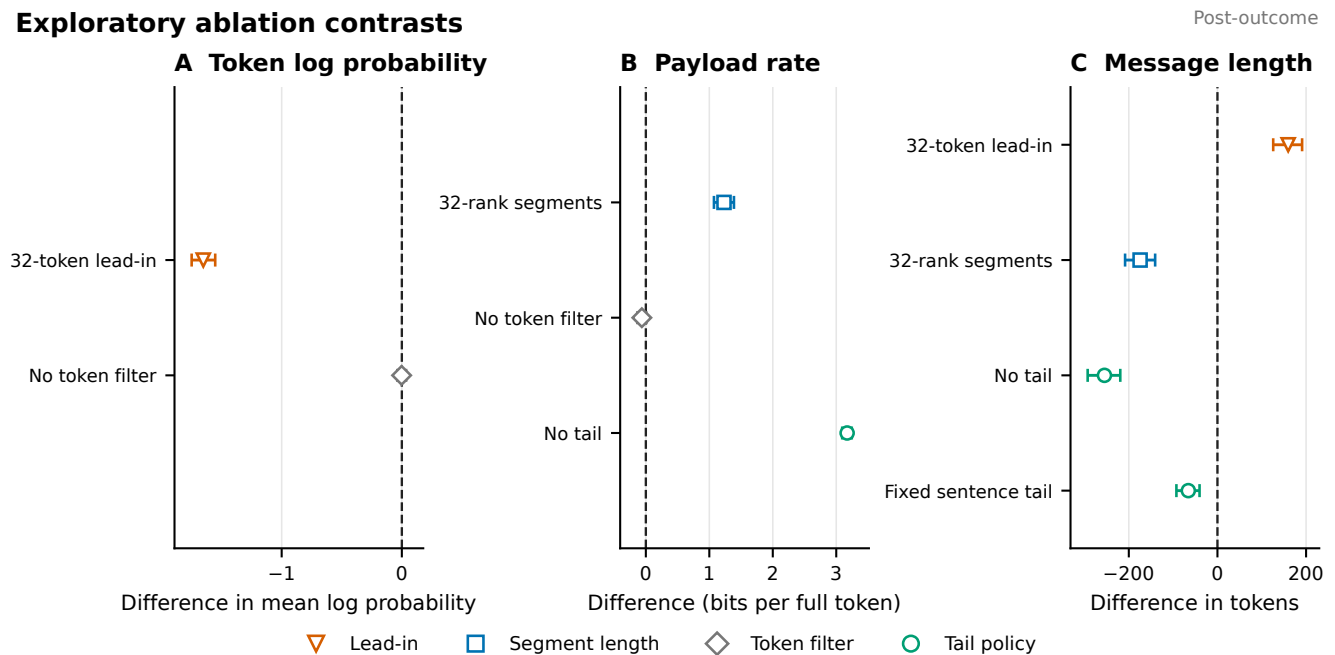


Figure S3. Ablation summary. Selected paired level-minus-canonical estimates are shown with payload-bootstrap 95% confidence intervals. Rows are exploratory/post-outcome evidence extraction from retained outputs. The Mistral roundtrip-stable-filter cell is unavailable for 48 work units; unavailable is not an execution failure. No-tail rate gains primarily reflect deletion of non-payload tokens and are not quality evidence.

Factor and level	Δ mean log probability (95% CI)	Δ full tokens (95% CI)	Δ bits/full token (95% CI)	Interpretation or availability
Lead-in: 2 tokens	-0.575880 [-0.667347, -0.491716]	+13.534722 [-21.431424, 50.179861]	-0.063224 [-0.157433, 0.024628]	Likelihood lower; length and rate intervals include zero
Lead-in: 4 tokens	-0.788830 [-0.883630, -0.694343]	+0.520833 [-36.346181, 35.365104]	-0.051850 [-0.130954, 0.029064]	Likelihood lower; length and rate intervals include zero
Lead-in: 8 tokens	-1.041886 [-1.115991, -0.961546]	+56.750000 [25.909201, 87.697396]	-0.179845 [-0.262452, -0.101276]	Lower likelihood, longer cover, lower rate
Lead-in: 16 tokens	-1.367867 [-1.450430, -1.285953]	+124.645833 [85.268229, 164.115799]	-0.299132 [-0.379449, -0.216430]	Lower likelihood, longer cover, lower rate
Lead-in: 32 tokens	-1.652327 [-1.749091, -1.553103]	+159.590278 [125.829861, 191.030035]	-0.388736 [-0.469659, -0.312450]	Largest adverse lead-in contrast
Segment size: 4 ranks	+0.116065 [0.045376, 0.192279]	+212.527778 [172.067535, 253.639236]	-0.373965 [-0.451966, -0.300845]	More segment tails; longer cover and lower rate
Segment size: 16 ranks	+0.064803 [-0.000267, 0.131660]	-104.694444 [-135.813021, -74.589410]	+0.524216 [0.418241, 0.640763]	Fewer tails; shorter cover and higher rate
Segment size: 32 ranks	+0.152306 [0.076589, 0.230426]	-174.500000 [-208.405035, -140.573264]	+1.231461 [1.073116, 1.388188]	Fewer tails; shortest selected cover contrast
No filter	-0.001797 [-0.037738, 0.035394]	+30.090278 [1.998264, 59.432986]	-0.062969 [-0.134265, 0.007651]	No clear likelihood or rate benefit
Stable filter	0.000000 [0.000000, 0.000000]	0.000000 [0.000000, 0.000000]	0.000000 [0.000000, 0.000000]	Two-model invariant; Mistral unavailable, not failed
No tail	0.000000 [0.000000, 0.000000]	-254.951389 [-292.821007, -219.249479]	+3.171551 [3.094005, 3.244380]	Mechanical deletion of non-payload tokens
Fixed sentence tail	0.000000 [0.000000, 0.000000]	-65.708333 [-92.590451, -40.372917]	-0.037921 [-0.113898, 0.033508]	Shorter than dynamic; rate interval includes zero

Table S10. Reviewer-relevant ablation contrasts. Differences are level minus the canonical setting (no lead-in, eight ranks per segment, safe-text filter, and dynamic tail). Estimates pool three available models and 48 paired payloads per model except the stable-filter row, which contains two models because the 48-unit Mistral condition was unavailable. Intervals are payload-bootstrap 95% intervals. Mean log probability is evaluated on payload-bearing forced positions; zero tail-policy differences arise because tail choice does not change those positions. All rows are exploratory/post-outcome.

Prompt-topic contrasts

The topic extraction compares the multi-topic and single-topic segmented schedules within model. Artifact counts are ratios on the response scale so the remaining estimates are multi-topic minus single-topic contrasts on their reported model scales. One of 15 rows excluded the null: the Llama expected artifact-count ratio was 1.276361 (95% CI, 1.114743 to 1.461409; Holm-adjusted $p = 0.003294$). Every other Holm-adjusted value was at least 0.396248. This locked extraction is exploratory and post-outcome where it supports neither a general topic advantage nor a new confirmatory hypothesis.

A separate exploratory subgroup display defines tail gain as full-message mean token log probability minus forced-span mean token log probability. It contains 8,280 paired segments from 1,440 segmented trials across three models, six prompt categories, and single-topic and multi-topic schedules, giving 36 displayed groups. Every group contains 230 segments, and all 36 group means were positive, ranging from +2.825721 to +3.652576. The numbers of independent payload clusters differ by schedule, 40 for single-topic and 190 for multi-topic. Uncertainty uses 2,000-resample payload-cluster bootstrapping stratified by payload class. This subgroup estimand is not equivalent to the adjusted single-topic versus multi-topic mixed-model contrast where the adjusted source-quality contrasts ranged from -0.02214 to +0.01881 and had Holm-adjusted $p = 1.0$ for all three models.

Tail gain by topic

230 segments/group; payload clusters: single-topic n=40, multi-topic n=190

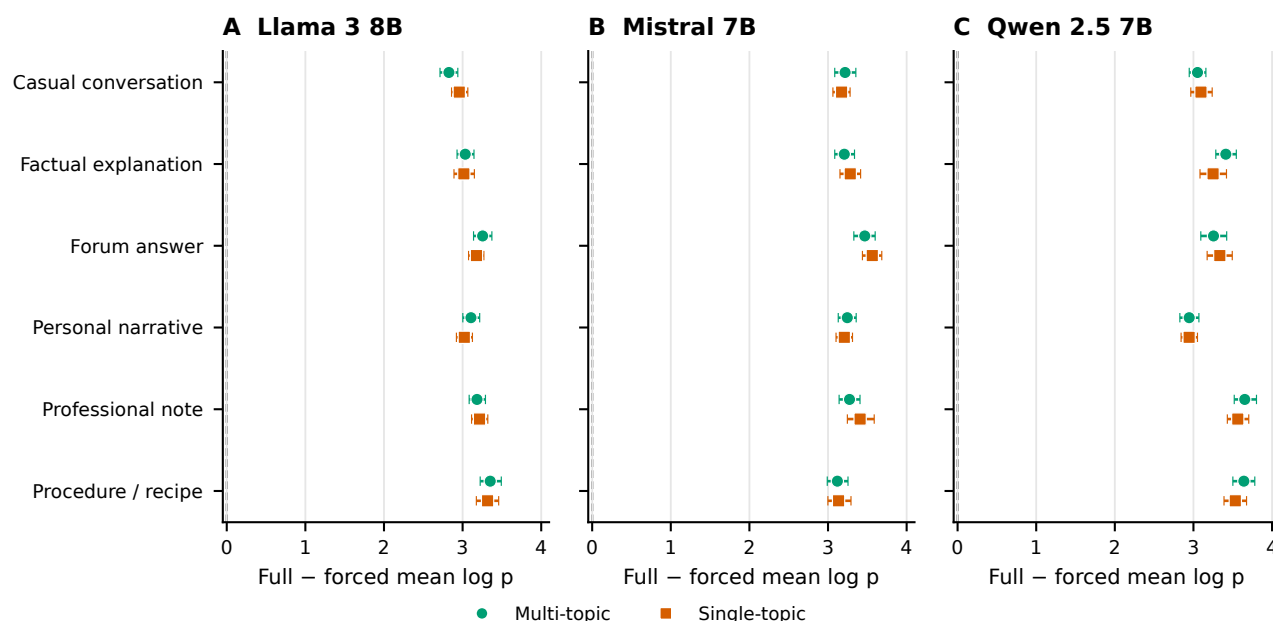


Figure S4. Exploratory tail gain by prompt category. Tail gain is full-message mean token log probability minus forced-span mean token log probability. The 8,280 paired segments arise from 1,440 segmented trials across three models, six prompt categories, and two topic schedules, yielding 36 displayed groups of 230 segments each. All group means are positive and range from +2.825721 to +3.652576. Uncertainty uses 2,000-resample payload-cluster bootstrapping stratified by payload class; independent payload-cluster counts are 40 for single-topic groups and 190 for multi-topic groups. This exploratory subgroup estimand is not the adjusted schedule contrast: adjusted source-quality contrasts range from -0.02214 to +0.01881, with Holm-adjusted $p = 1.0$ for all three models.

Table S11. All 15 prompt-topic contrasts.

Outcome	Model	Estimate	95% CI	p_{Holm}
Primary artifact count	Llama	1.276361	[1.114743, 1.461409]	0.00329358
Primary artifact count	Mistral	1.015189	[0.848773, 1.214233]	1
Primary artifact count	Qwen	0.957147	[0.781516, 1.172247]	1
Source-model log probability	Llama	-0.022140	[-0.118718, 0.074439]	1
Source-model log probability	Mistral	0.016347	[-0.080231, 0.112925]	1
Source-model log probability	Qwen	0.018813	[-0.077766, 0.115391]	1
Effective artifact rate	Llama	0.012473	[-0.075124, 0.100070]	1
Effective artifact rate	Mistral	0.006792	[-0.080805, 0.094389]	1
Effective artifact rate	Qwen	-0.015822	[-0.103419, 0.071775]	1
Held-out evaluator log probability	Llama	-0.012801	[-0.129427, 0.103825]	1
Held-out evaluator log probability	Mistral	0.012046	[-0.104580, 0.128671]	1
Held-out evaluator log probability	Qwen	0.024711	[-0.091915, 0.141336]	1
Payload throughput	Llama	0.018304	[-0.016147, 0.052755]	0.595439
Payload throughput	Mistral	0.026471	[-0.007981, 0.060922]	0.396248
Payload throughput	Qwen	-0.009378	[-0.043829, 0.025073]	0.595439

Supplementary Note S7: Historical V2 neural steganalysis and leakage sensitivity

Corpus, architectures, and splits

This note preserves the historical V2 detector benchmark for continuity and work-unit provenance; its 15,840-row splits and results are not the current detector estimates and are superseded by the strictly deduplicated V3 analysis in Supplementary Notes S11–S12. The historical corpus contains 7,920 RankCloak covers and 7,920 matched ordinary controls, grouped by 480 payloads. Payload groups, rather than rows, were assigned to train, validation, and test sets. The TextCNN-equivalent implementation lowercases and tokenizes text into a 30,000-term vocabulary, truncates or pads to 256 tokens, uses 300-dimensional learned embeddings and 100 filters at each of widths 3 and 5, applies dropout 0.5, and optimizes with Adam at learning rate 0.001. The pretrained DeBERTa-v3-base classifier uses maximum length 256, learning rate 2×10^{-5} , weight decay 0.01, and three epochs. For each architecture, the frozen design comprises one matched split, 18 held-out-template splits, three leave-one-model splits, and six leave-one-codec splits. Thus 28 fits per architecture and 56/56 fits overall completed, producing 55,440 prediction rows. No row-level reassignment was used to improve performance.

For this historical benchmark, the reported endpoints were ROC-AUC and balanced accuracy. Precision, Brier score, and true-positive rate at false-positive rate at most 1% were supplementary/exploratory post-freeze measures. All confidence intervals for an individual split use 2,000 payload-group bootstrap resamples. Ranges across heterogeneous held-out splits are descriptive minima and maxima, not confidence intervals. Every historical split-level estimate and interval remains in the machine-readable supplementary CSV (`supplementary_tables/detector_split_metrics.csv`).

Neural detectability

Higher = easier detection · D-F exploratory

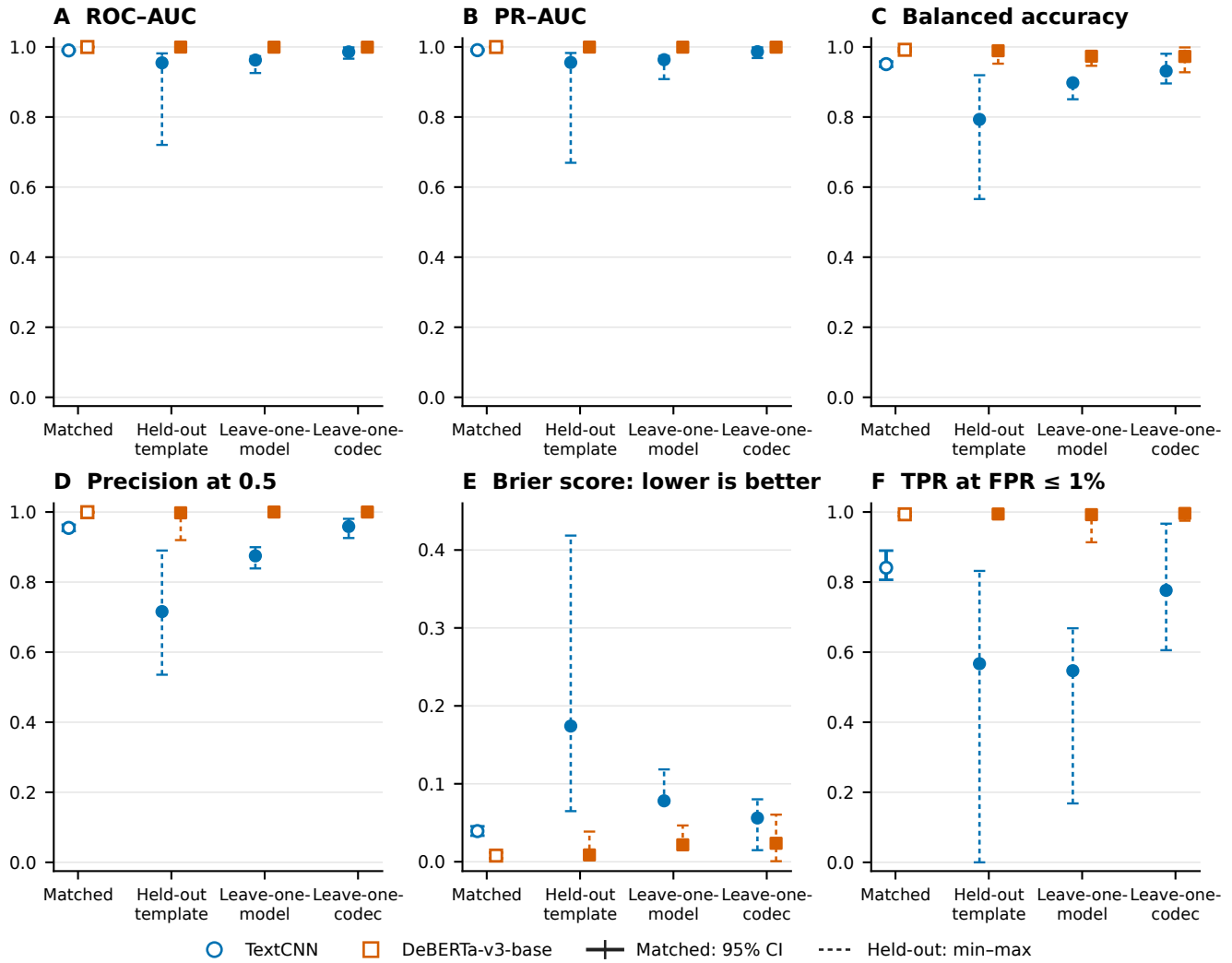


Figure S5. Historical V2 neural-detector evaluation. The six panels preserve the earlier frozen endpoints and supplementary exploratory metrics; current strictly deduplicated estimates are in Supplementary Note S12. Matched error bars are payload-group bootstrap 95% confidence intervals. Held-out regime bars span the minimum and maximum across prespecified splits and are not confidence intervals. Lower Brier scores are better; high values in the other panels indicate stronger distinguishability and therefore weaker concealment.

Architecture	Matched metric	Estimate	Payload-bootstrap 95% CI	Status
TextCNN-equivalent	ROC-AUC	0.990190	[0.987740, 0.992682]	Historical V2 frozen
TextCNN-equivalent	Balanced accuracy	0.951010	[0.943456, 0.958466]	Historical V2 frozen
TextCNN-equivalent	Precision	0.954684	[0.945891, 0.963421]	Supplementary exploratory
TextCNN-equivalent	Brier score	0.039320	[0.033497, 0.045453]	Supplementary exploratory
TextCNN-equivalent	TPR at FPR $\leq 1\%$	0.840909	[0.806517, 0.889752]	Supplementary exploratory
DeBERTa-v3-base	ROC-AUC	0.999834	[0.999696, 0.999939]	Historical V2 frozen
DeBERTa-v3-base	Balanced accuracy	0.991919	[0.987654, 0.995665]	Historical V2 frozen
DeBERTa-v3-base	Precision	0.999487	[0.998393, 1.000000]	Supplementary exploratory
DeBERTa-v3-base	Brier score	0.007895	[0.004432, 0.011787]	Supplementary exploratory
DeBERTa-v3-base	TPR at FPR $\leq 1\%$	0.993434	[0.988724, 0.998387]	Supplementary exploratory

Table S12. Historical V2 matched detector endpoints, superseded for current claims by Supplementary Note S12. Higher ROC-AUC, balanced accuracy, precision, and low-FPR sensitivity are adverse for concealment; lower Brier score indicates better detector calibration.

Architecture	Regime (split count)	ROC-AUC range	Balanced-accuracy range
TextCNN-equivalent	Held-out template (18)	[0.720421, 0.981705]	[0.565909, 0.919318]
TextCNN-equivalent	Leave one model (3)	[0.925623, 0.973650]	[0.850758, 0.883333]
TextCNN-equivalent	Leave one codec (6)	[0.966532, 0.998634]	[0.895833, 0.980556]
DeBERTa-v3-base	Held-out template (18)	[0.996291, 0.999979]	[0.952273, 0.995455]
DeBERTa-v3-base	Leave one model (3)	[0.996772, 0.999979]	[0.945455, 0.970455]
DeBERTa-v3-base	Leave one codec (6)	[0.997917, 1.000000]	[0.927778, 0.998611]

Table S13. Historical V2 ranges across heterogeneous frozen held-out splits. These ranges are not pooled confidence intervals; current leave-one-family-out results are in Supplementary Note S12.

Leakage, exclusion sensitivity, repeatability, and cost

In the historical V2 benchmark, exact train/test visible-text identity and exact tokenizer-sequence identity checks passed in all 28 frozen split definitions. A separate normalized character n -gram diagnostic found 53 train/test near-duplicate pairs, of which 44 crossed payload groups, affecting 16 splits. Excluding affected test payload groups without retraining changed ROC-AUC from -0.001925 to +0.004568 for TextCNN-equivalent and from -0.000038 to +0.000166 for DeBERTa-v3-base. Because that sensitivity did not rebuild training data, it motivated and is superseded by the pre-feature component construction and zero-linkage audit in Supplementary Note S11.

Fixed-seed reruns on the same CUDA device produced exact metric equality for both architectures. This is evidence of same-device repeatability only. CPU neural training was infeasible in the retained run and no CPU result was substituted; the study makes no CPU/GPU equivalence claim. The representative CUDA benchmark recorded TextCNN wall/training times 91.358113/9.497936 seconds and peak allocated VRAM 306,247,680 bytes; DeBERTa values were 1,536.660910/1,425.063414 seconds and 6,380,851,200 bytes.

Supplementary Note S8: Capacity and quality theory

The main Methods defines bounded forced length, nominal and effective rates, full-message overhead, and the same-context probability bound. This note gives finite-code consequences, proof conditions, and the retained-field audit without repeating those central equations.

Finite code space and exact inversion

Let H , B , and n_B have the meanings in Main Eq. (14). The unused capacity of the smallest fixed-length rank word is described by

$$\sigma_B = n_B \log_2 B - H, \quad \eta_B = \frac{2^H}{B^{n_B}} = 2^{-\sigma_B}, \quad u_B = 1 - \eta_B.$$

Here σ_B is code-space slack in bits, η_B is utilization, and u_B is the unused codeword fraction. For a power-of-two alphabet and an integer-length bitstream, σ_B equals literal terminal padding bits. The byte codec records those bits explicitly; for $B = 8$, one

byte is the finite example $n_B = 3$ with one terminal padding bit, while a multi-byte payload is packed as one continuous stream. Hex nibbles have zero slack when the represented source length is exactly four bits per canonical character. The rank-bound proposition assumes that every forced history has at least B eligible candidates and that encoder and decoder reproduce the same model, token IDs, prompt context, mask, and ordering. A digit $d \in \{0, \dots, B-1\}$ selects the token at filtered rank $d+1$; exact replay ranks that same saved token as $d+1$. Serial induction over the forced span recovers the rank word, and injectivity of the codec recovers the representation. The proposition does not apply after an altered token, context, boundary, mask, or model, and it supplies no edit-correction property.

Segmentation and overhead decomposition

For segment forced lengths s_j , lead-in lengths ℓ_j , and tail lengths t_j , the saved records satisfy $\sum_j s_j = n_B$. The empirically audited residual can be decomposed as

$$D = N_{\text{full}} - n_B = \sum_{j=1}^J (\ell_j + t_j) + \delta, \quad \delta \geq 0.$$

Thus segmentation alone preserves forced-position count, while more resets can create more opportunities for non-payload continuations. A no-tail condition raises bits per full token by shrinking the denominator. This is an arithmetic consequence and not evidence that the resulting visible text is more readable or natural.

Conditional likelihood gap

The per-history ordering in Main Eq. (16) permits a trace-level comparison with the rank-1 eligible token and, for bounded codecs, the rank- B token. In addition to Q_B , the audit records

$$\Delta_B = Q_B - Q_{\text{greedy}} = \frac{1}{n_B} \sum_{i=1}^{n_B} [-\log p_M(y_i | h_i) + \log p_{(1)}(h_i)] \geq 0.$$

The upper rank- B endpoint is evaluated only when its probability was retained under the identical filtered history. Direct subword traces have no single fixed B , so that endpoint is unavailable rather than imputed. These quantities are model-scale surprisals in nats per forced token, not linguistic or participant outcomes.

Empirical checks and trace boundary

The validation inventory contained 17,424 eligible retained records. Of these, 7,008 had all fields required for complete capacity, rate, and supported quality-bound checks. Across evaluable traces there were zero forced-position-count violations, zero rank/rate-bound violations, and zero supported quality-bound violations. The nine aggregate cells in the theoretical capacity panel of Supplementary Fig. S6 lay exactly on the identity between theoretical and observed forced positions. There were 2,976 positive cover-length residuals, attributable to natural tails or other counted cover extension.

The complete encoder decoder cascade proposition is not directly trace-evaluable from the saved record structure: none of the retained rows simultaneously carries every intermediate object required to replay the entire proof chain as one trace. The algebraic results are therefore conditional proofs and the retained-field checks are supporting empirical validation, not a complete end-to-end trace proof. They also do not validate human naturalness.

This theoretical validation asks whether the retained fields satisfy the declared fixed-radix identities, rate bounds, and supported same-context likelihood bound. It is distinct from the empirical capacity and quality frontier in Main Fig. 2, which compares observed representation rates and automated model/surface diagnostics on common payload support.

Capacity and tail overhead

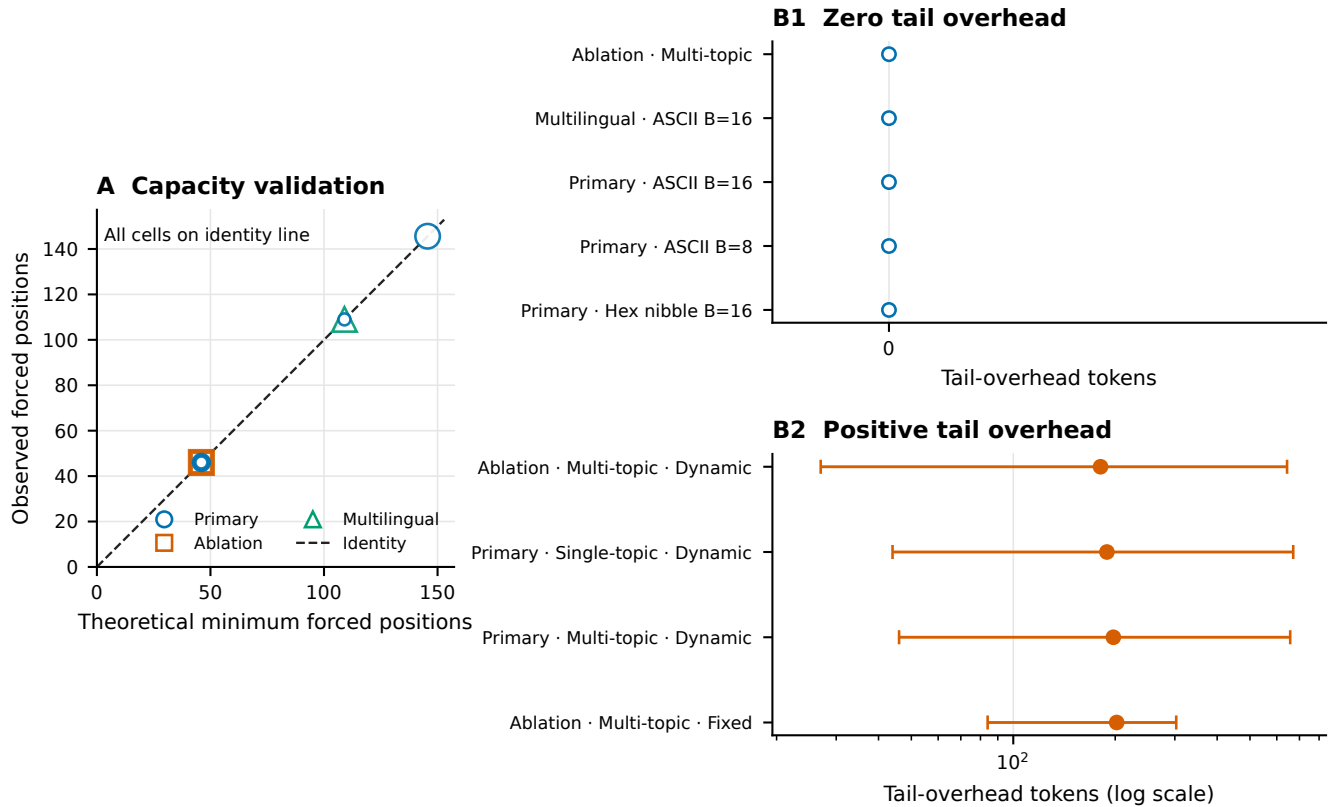


Figure S6. Theoretical capacity and tail validation. Fixed-radix payload length and effective full-message rate are checked against their declared algebraic bounds, while the cover-length residual isolates tokens beyond the forced payload positions. Across 17,424 retained records, 7,008 were evaluable for the complete plotted capacity and supported quality-bound checks; no forced-position, rate-bound, or evaluated quality-bound violation occurred. Positive residuals in 2,976 records represent natural tails or other cover overhead. This validation tests retained-field identities and differs from the empirical common-support frontier in Main Fig. 2; neither analysis is evidence of human-perceived naturalness.

Validation item	Result	Interpretation
Eligible records	17,424	Broad retained validation inventory
Fully evaluable records	7,008	All required trace fields present
Forced-position violations	0	Observed count matched codec requirement
Rank/rate violations	0	Evaluated bounded-codec relations held
Supported quality-bound violations	0	Model-scale trace bound held where evaluable
Positive cover-length residuals	2,976	Tail/cover overhead, not error
Complete cascade traces	0	End-to-end proposition not directly trace-evaluable

Table S14. Empirical capacity and traceability audit.

Supplementary Note S9: Computational overhead

Primary-stage timing used 12,144 matched payload rows across the 18 model and protocol cells in Table S15. Intervals are 2,000-resample payload-group bootstrap intervals. Generation time is retained inclusive wrapper wall time; representation, filter setup, model generation, and supported recovery fields are reported separately in the sealed tables but are not claimed to be perfectly isolated kernel timings. Encoding wrapper time and supported decoding measurements consequently should not be summed as independent hardware primitives.

Across the 18 cells, mean inclusive generation time ranged from 1.674368 to 16.087628 seconds and payload throughput from 13.631855 to 100.851772 bits/s; effective payload rate ranged from 0.710004 to 7.322003 bits per full token. Direct-rank

and hex-nibble protocols were generally faster than the byte-expanded conditions, while segmented protocols were generally slower because they repeated prompt and tail work. These retained inclusive wall-time measurements support computational accounting under the recorded hardware and harness, not universal hardware-performance claims.

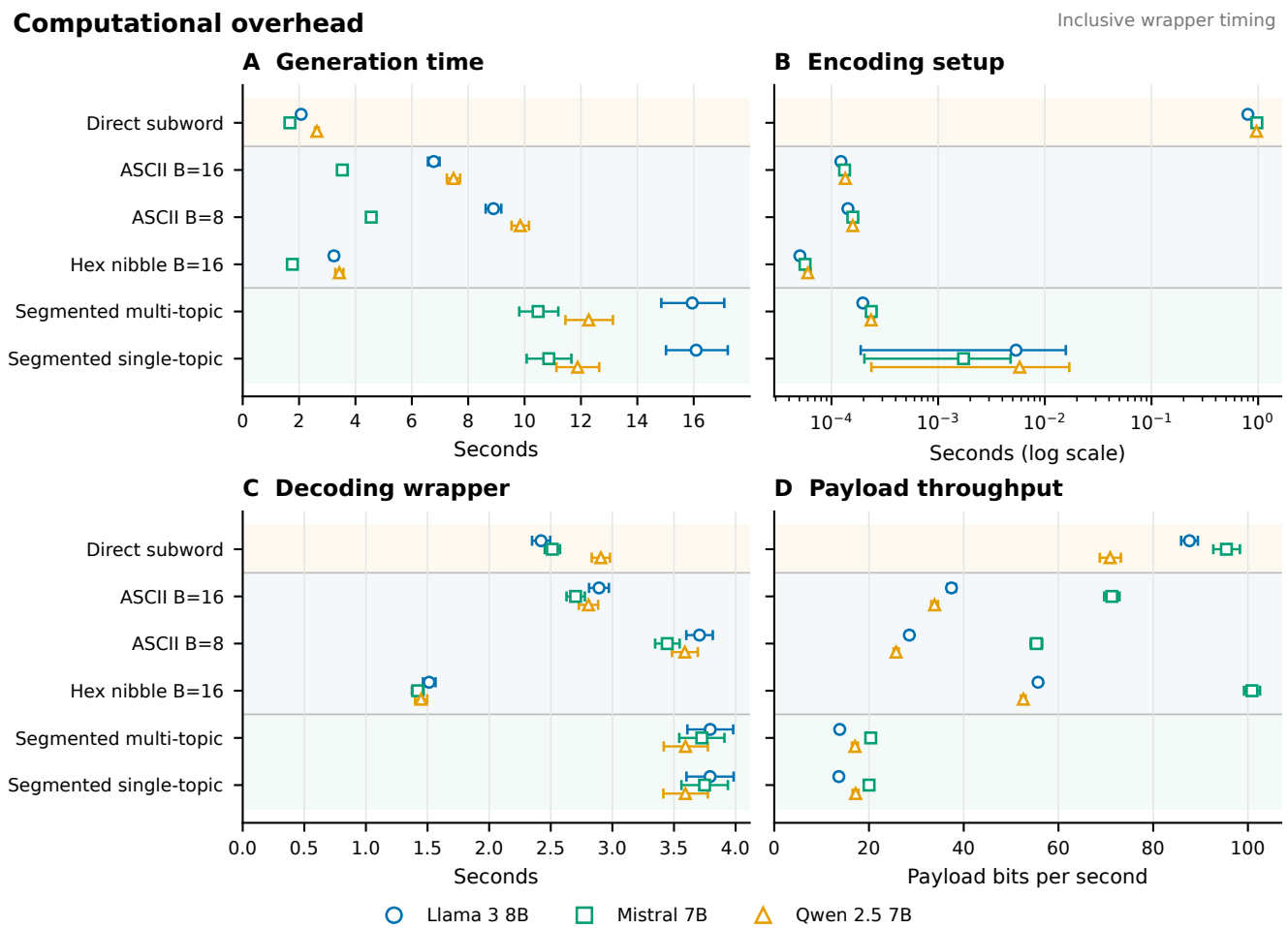


Figure S7. Complete computational overhead summary. Primary model and protocol cells show payload-bootstrap summaries for inclusive generation, encoding setup, supported recovery, throughput, and effective rate. The setup panel uses a logarithmic axis because cells span several orders of magnitude. CPU time and repeated warm-up measurements were unavailable and wall time is not relabeled as CPU time.

Model	Protocol	<i>n</i>	Mean generation time, s (95% CI)	Mean payload throughput, bit/s (95% CI)
Llama	Direct ranks	480	2.074349 [2.014282, 2.135219]	87.658980 [85.892805, 89.433262]
	ASCII-16	480	6.776682 [6.572039, 6.982913]	37.409987 [36.542507, 38.322129]
	ASCII-8	480	8.896445 [8.619910, 9.172562]	28.522326 [27.867278, 29.213688]
	Hex-16	240	3.236699 [3.099398, 3.376963]	55.691381 [55.148301, 56.225745]
	Segmented multi-topic	240	15.946543 [14.850856, 17.084494]	13.827671 [13.155163, 14.496873]
	Segmented single-topic	240	16.087628 [15.017670, 17.208756]	13.631855 [12.954794, 14.325096]
Qwen	Direct ranks	480	2.632108 [2.565827, 2.699101]	70.911677 [68.729360, 73.171915]
	ASCII-16	480	7.480319 [7.242905, 7.714463]	33.843456 [33.153343, 34.571271]
	ASCII-8	480	9.847052 [9.537974, 10.155572]	25.724469 [25.198423, 26.278772]
	Hex-16	240	3.427555 [3.284745, 3.575041]	52.584782 [52.074452, 53.086870]
	Segmented multi-topic	240	12.276814 [11.449256, 13.133478]	17.040206 [16.420206, 17.655385]
	Segmented single-topic	240	11.881962 [11.129651, 12.646529]	17.182540 [16.534412, 17.783251]
Mistral	Direct ranks	480	1.674368 [1.637117, 1.711610]	95.454287 [92.689931, 98.318820]
	ASCII-16	480	3.532546 [3.430360, 3.637370]	71.264443 [69.655729, 72.892828]
	ASCII-8	480	4.558496 [4.421249, 4.699172]	55.340577 [54.149134, 56.561483]
	Hex-16	240	1.762009 [1.698523, 1.826194]	100.851772 [99.168287, 102.554774]
	Segmented multi-topic	240	10.478402 [9.810233, 11.196055]	20.368846 [19.484536, 21.270824]
	Segmented single-topic	240	10.858662 [10.074554, 11.660671]	20.014386 [19.095928, 20.969663]

Table S15. Inclusive generation time and payload throughput by model and protocol. Llama denotes Meta-Llama-3-8B-Instruct, Qwen denotes Qwen2.5-7B-Instruct, and Mistral denotes Mistral-7B-Instruct-v0.3; all use the reported Q4_K_M configurations. Brackets contain payload-bootstrap 95% confidence intervals.

Detector	Wall time (s)	Training time (s)	Peak allocated VRAM	Scope
TextCNN-equivalent	91.358113	9.497936	306,247,680 bytes	One fixed CUDA benchmark
DeBERTa-v3-base	1,536.660910	1,425.063414	6,380,851,200 bytes	One fixed CUDA benchmark

Table S16. Representative detector cost. CPU time, repeat warm-up measurements, and cross-device benchmarks were unavailable.

Supplementary Note S10: Worked examples and claim-boundary audit

Representative retained examples

Payload to ranks to cover. A retained Llama segmented-hex trial used the public SHA-256 test-vector serialization

7cec7f1e68bbef4fdf066a51ccbea4377ee369e1b53219743aa386615e12d44a.

The first eight encoded ranks were [8, 13, 15, 13, 8, 16, 2, 15]. Under the saved prompt/model/filter state, the associated forced token IDs were [1789, 198, 16, 627, 13617, 10652, 25, 9842], detokenizing to the deliberately mechanical span

For\nl.\nExample conversation: Write

The subsequent visible text

a friendly message to a friend about making relaxed plans for the weekend.

was a natural tail. It carried no payload rank and was ignored by the decoder. Saved-token replay reproduced all nibble ranks and the serialized digest exactly. This example illustrates deterministic rank realization, not an assertion that the forced span is human-natural.

Visible-text retokenization failure. In a retained ablation trial, detokenization followed by tokenizer reapplication first diverged in segment 6, forced position 5, expected token ID 11,623 at rank 8 became observed token ID 53,992 at replay rank 2. The visible string alone therefore did not preserve the token boundary needed by the saved-token contract.

255 **Transformation failure.** For the SHA-256 example above, straight-to-smart quote conversion failed at segment 1, position 1: expected token ID 1 at rank 9, while the transformed replay observed token ID 2,118 at rank 155. The first-divergence record localizes the mismatch but does not prove that a single changed glyph alone caused every later error.

Claim-boundary audit

Table S17. Claim boundaries applied throughout the revision.

Issue	Supported statement	Prohibited or unresolved extension
Exact-copy recovery	6,480/6,480 primary serialized payloads recovered from saved token IDs under matched replay	Ordinary visible-text reliability, cross-model robustness
Transmission	Controlled frozen perturbations reveal severe fragility; tail-only truncation preserves recovery when only ignored tokens are removed	Broad edit robustness or general mitigation
Human evaluation	504 computational stimuli have automated surface/model diagnostics	Human-perceived readability or naturalness; zero participants and ratings
Deployment	No operational channel or deployment traffic was tested	Practical covert communication or real-world validation
Detectability	Two data-adaptive text-only detectors and an exact-model-aware attacker strongly distinguish evaluated covers; high AUC is adverse	Undetectability or resistance to future detectors
Multilingual scope	288/288 Spanish and 288/288 Simplified Chinese exact-copy secondary trials recovered	Broad multilingual naturalness or cross-language generalization
Unavailable work	432 predeclared combinations lacked compatible retained outputs	Calling unavailable combinations experimental failures
Model replay	Retained outputs are bound to model/configuration manifests	Fresh local generation replay while configured GGUF files are absent
Detector leakage	Strict pre-feature component split has zero audited cross-partition links	Semantic paraphrases below the declared 0.95 threshold are not excluded
Statistical models	Reported conditional estimates retain diagnostic warnings	Treating three singular mixed-model fits as unqualified evidence; substituting an undeclared fallback
Exploratory analyses	Compact ablation, topic, and auxiliary detector analyses are labeled post-outcome/exploratory where applicable	Recasting them as preregistered confirmatory evidence
Overhead	Retained inclusive wall-time and fixed CUDA benchmark scopes are reported	CPU time, warm-up stability, CPU/GPU equivalence, or hardware-general performance
Theory	Conditional rank/capacity relations and retained-field checks are supported	A directly trace-evaluated full cascade or human-quality proof
Cryptography	The carrier can encode outputs from declared cryptographic algorithms	Cryptographic security supplied by RankCloak itself; confidentiality without an external cryptosystem

Supplementary Note S11: Strict detector splits and human-authored secondary controls

Normalization, deduplication, and partition audit

All V3 detector corpora were deduplicated before feature extraction or tokenization. Visible text underwent Unicode NFKC normalization, case folding, Unicode-whitespace collapse, and SHA-256 hashing. If a normalized exact duplicate implicated either member of a RankCloak/control pair, the complete matched pair was removed so that class balance and pairing were preserved. Near duplicates were defined before splitting as cosine similarity at least 0.95 under L2-normalized character-boundary TF-IDF with 3–5 character n -grams and minimum document frequency two. Payload groups connected by any such edge formed immutable components.

One normalized exact-duplicate group implicated two rows in the 15,840-row source corpus. Removing that complete matched pair left 15,838 observations. The remaining audit found 52 near-duplicate pairs, including 44 that crossed payload groups. A deterministic seed-20260831 balancing procedure assigned whole connected components to train, validation, and test. The final audit found zero payload, matched-pair, normalized-text, component, or declared near-duplicate links between partitions (Table S18). The statement of zero leakage is limited to these declared identities and thresholded edges; it cannot rule out semantic paraphrases below the threshold or guarantee real-world generalization.

Audit item	Count	Interpretation
Source observations	15,840	Balanced RankCloak and matched ordinary-model controls
Normalized exact groups	1	One complete matched pair removed, two rows total
Retained observations	15,838	9,904 train; 2,952 validation; 2,982 test
Declared near-duplicate pairs	52	44 cross-payload; all connected groups kept in one partition
Audited cross-partition links	0	Zero for payload, pair, normalized text, component, and declared near duplicate

Table S18. Strict detector-corpus construction. Counts are from the locked V3 deduplication audit. Zero audited leakage applies only under the stated normalization and similarity rule.

Human-authored control selection and scope

The secondary controls came from Databricks Dolly 15k v1.0, revision bdd27f4d94b9c1f951818a7da7fd7aeaa5dbff1a, under CC BY-SA 3.0. Deterministic screening retained responses without automatic quality, personally identifiable information, or unsafe-content flags. The same exact and near-duplicate rules were applied within the human-authored candidates and against the primary corpus. Hungarian assignment matched candidates to RankCloak test texts within available prompt templates using relative word-count plus weighted character-count difference; relative word-count differences above 0.35 were rejected.

The pipeline selected 250 texts spanning 16 of 18 templates. It found no normalized exact duplicates, no within-human near-duplicate pairs, and no declared near-duplicate links to the primary corpus. Human labels never entered model fitting or threshold selection. The corpus is a bounded secondary computational control, not recruitment, a participant study, a user study, a human rating exercise, or proof of naturalness. Coverage is sparse or absent for several conversational, narrative, and professional templates, and detector performance can include generic machine-versus-human differences.

Human-control item	Result	Scope
Selected responses	250	Frozen test negatives from Dolly; 16 templates represented
Within-human exact links	0	Under NFKC/casefold/whitespace normalization
Within-human near links	0	Under the declared 0.95 character- n -gram rule
Human-to-primary near links	0	Same declared rule
Labels used for fitting/thresholds	No	Thresholds came from matched ordinary-model validation controls
Empirical 0.1% FPR	Unavailable	Only 250 human negatives; at least 1,000 required

Table S19. Human-authored secondary-control design and audit.

Supplementary Note S12: Deduplicated, unseen-family, model-aware, and low-FPR detection

The text-only attackers reuse the fixed V1 TextCNN and pinned microsoft/deberta-v3-base revision 8ccc9b6f36199bec6961081d44eb72fb3f7353f3. Their architectures and hyperparameters were not tuned on V3 test data. They are data-adaptive black-box attackers because they are fitted specifically to distinguish RankCloak from matched ordinary-model

290 output. The model-aware logistic attacker instead receives saved token log probabilities from the exact generation model and uses summary surprisal features. Its L2 regularization value was chosen from 0.01, 0.1, 1, and 10 by validation ROC-AUC. Rank-only fields were excluded because ordinary-control ranks were not stored.

295 Matched fitting used 9,904 training and 2,952 validation observations; the test had 1,491 positives and 1,491 negatives. For each leave-one-family-out fit, the target family was absent from both training and validation, and evaluation used only its 497 positive and 497 negative observations in held test components. Table S20 reports the adverse matched results and the range across the three genuine family holdouts.

Detector	Matched ROC-AUC (95% CI)	Matched normalized 0–1% pAUC	Leave-one-family-out ROC-AUC range
TextCNN	0.9868 [0.9819, 0.9913]	0.6652	[0.9186, 0.9784]
DeBERTa-v3-base	0.9994 [0.9984, 1.0000]	0.9948	[0.9972, 1.0000]
Model-aware surprisal	0.999995 [0.999970, 1.000000]	0.999460	[0.999943, 1.000000]

Table S20. Detection on strictly deduplicated components. The matched test has 1,491 observations per class. Ranges summarize three distinct held-family tests and are not pooled confidence intervals. Higher values are adverse for concealment.

300 ROC-AUC used tie-aware empirical ranks. The normalized 0–1% partial AUC was the exact empirical false-positive-budget step envelope divided by 0.01; no score or ROC vertex was interpolated. At target α , the validation threshold maximized TPR subject to no more than $\lfloor \alpha n_{\text{neg}} \rfloor$ false positives, with conservative tie breaking. The threshold was then frozen for test. A target was reported only if validation and test both contained at least $1/\alpha$ negatives. Confidence intervals used 2,000 percentile bootstrap resamples of complete deduplication clusters.

Detector	Evaluation	Val. negatives	Test negatives	Test TPR/FPR at frozen 1% threshold	Exact false positives	Frozen 0.1% result
TextCNN	Matched	1,476	1,491	0.8317/0.0127	19/1,491	TPR 0.4829; FPR 0.0013; 2/1,491 FP
DeBERTa	Matched	1,476	1,491	0.9913/0.0007	1/1,491	TPR 0.9886; FPR 0.0007; 1/1,491 FP
Model-aware	Matched	1,476	1,491	0.9987/0.0020	3/1,491	TPR 0.9987; FPR 0.0020; 3/1,491 FP
TextCNN	Held family range	984	497	TPR 0.7103–0.8330; FPR 0.0201–0.0704	10–35/497	Unavailable: fewer than 1,000 negatives
DeBERTa	Held family range	984	497	TPR 0.9759–1.0000; FPR 0.0040–0.0563	2–28/497	Unavailable: fewer than 1,000 negatives
Model-aware	Held family range	984	497	TPR 0.9920–1.0000; FPR 0.0000–0.0060	0–3/497	Unavailable: fewer than 1,000 negatives

Table S21. Validation-frozen low-FPR operating points. The observed test FPR can differ from the validation target. Exact counts, rather than interpolated rates, expose calibration drift and empirical resolution.

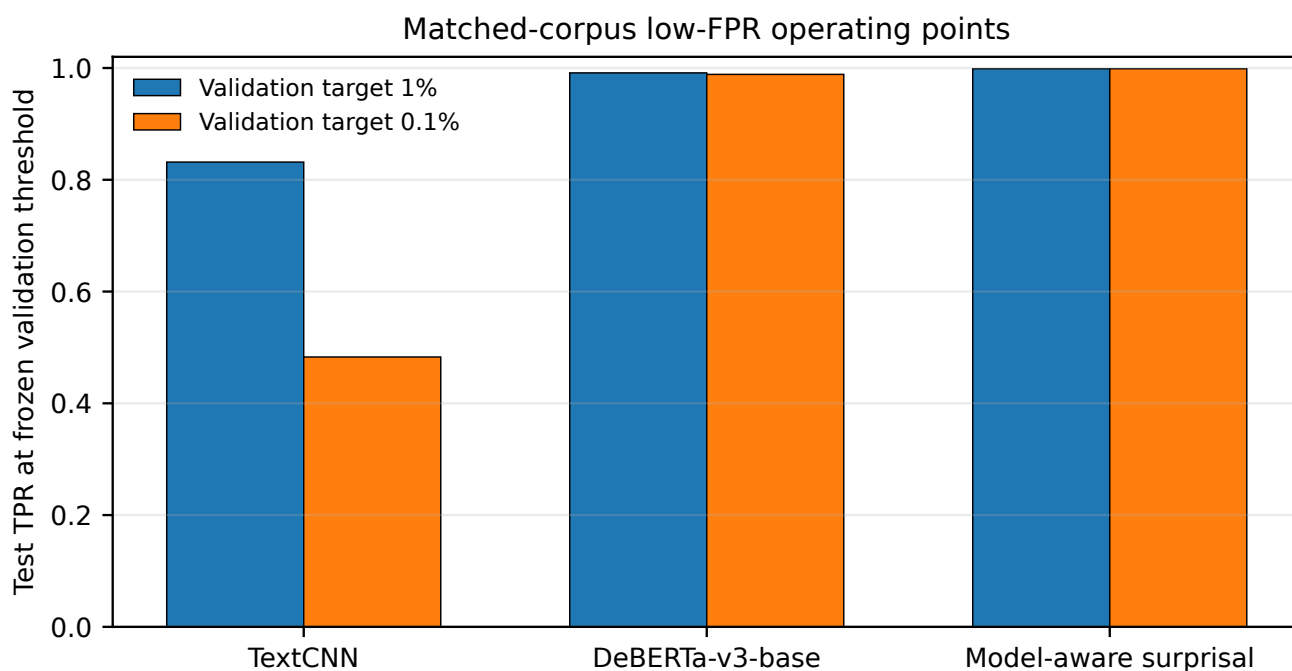


Figure S8. Low-FPR detection under frozen thresholds. Thresholds were selected on validation and applied once to test. Missing 0.1% family-holdout results reflect insufficient negative counts, not zero sensitivity.

Against the human-authored secondary controls, TextCNN ROC-AUC was 0.7537 (95% CI [0.7141, 0.7918]) and DeBERTa ROC-AUC was 0.9846 (95% CI [0.9771, 0.9910]). At the thresholds selected on ordinary-model validation controls for nominal 1% FPR, TextCNN classified 152/250 human texts as positive (FPR 0.6080; grouped-bootstrap 95% CI [0.5469, 0.6667]) and DeBERTa classified 97/250 as positive (FPR 0.3880; CI [0.3263, 0.4466]). These high false-positive rates show domain sensitivity and do not reverse the matched-control result.

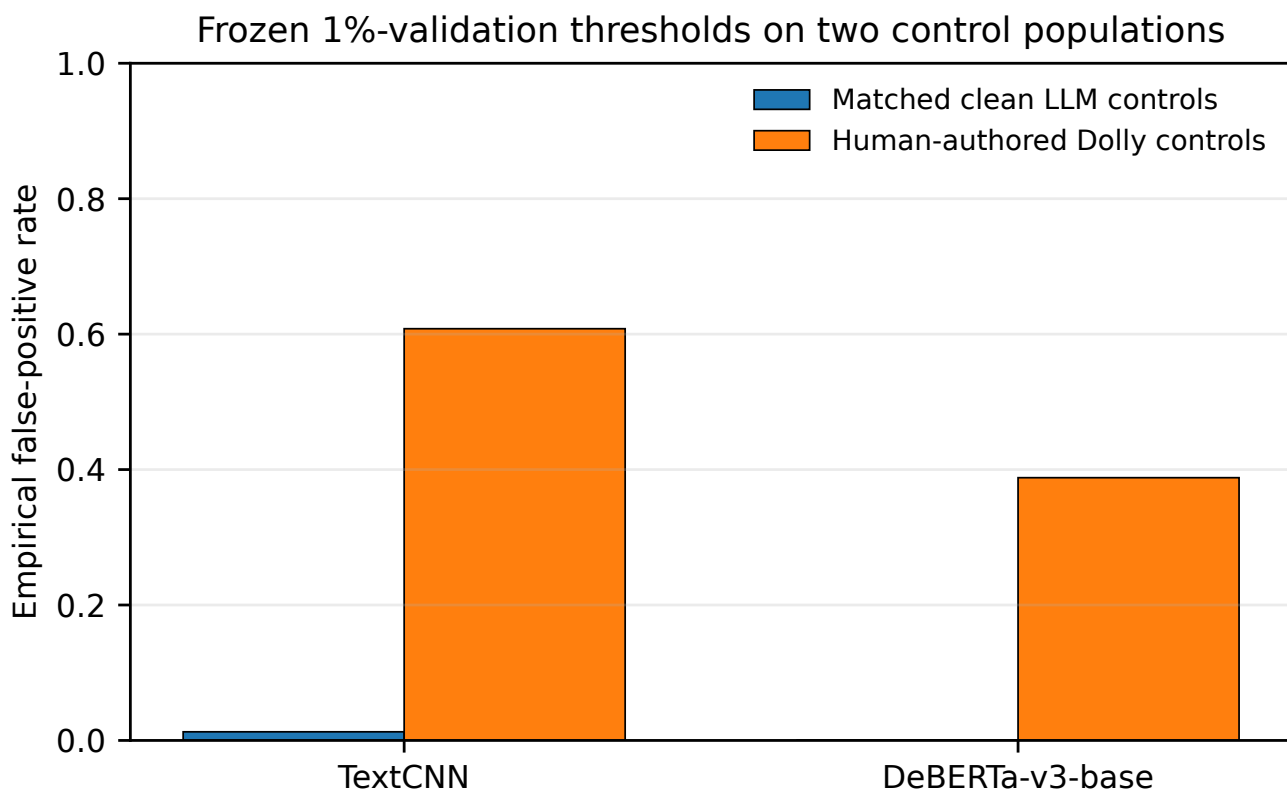


Figure S9. Secondary human-authored control evaluation. Human-authored labels were not used for fitting or threshold selection. The comparison may include general machine-versus-human signal and is not a human evaluation.

The two learned detectors adapt to the observed training distribution; the likelihood-based detector represents an attacker with exact model access. Their strength makes the negative detectability conclusion more credible within the evaluated corpus, but none is a comprehensive adversarial search over all classifiers, prompts, post-processing, models, or deployment distributions.

Supplementary Note S13: Entropy-gated payload embedding

Calibration and replay-consistent gate

The entropy-gated extension was informed by Cai, Ding, and Tao¹, but the objectives differ. Their watermarking detector decides whether text carries a mark; a RankCloak receiver must reconstruct an arbitrary payload. Selective watermark insertion therefore does not by itself establish payload capacity, recoverability, or concealment.

Six 128-position ordinary top- p traces per model used temperature 0.8, top- p 0.95, and the frozen serial sampler, yielding 768 development positions per model. Detector outcomes did not enter calibration. The median threshold defines the moderate gate and the 75th percentile defines the strict gate (Table S22).

Model	Traces	Median threshold, bit	75th-percentile threshold, bit
Llama 3 8B Q4_K_M	6	0.803259	1.705455
Mistral 7B Q4_K_M	6	0.966861	1.917551
Qwen 2.5 7B Q4_K_M	6	1.361919	2.432557

Table S22. Frozen entropy thresholds from ordinary-generation development traces.

Before each embedding-span step, the encoder computed filtered next-token Shannon entropy. An eligible position consumed and forced the next payload rank. An ineligible position used ordinary top- p sampling under the same allowed-token mask and did not consume a payload rank. Replay recomputed entropy before each observed token, consumed a rank only at

eligible positions, ignored observed ranks at sampled skips, and appended every observed token to context. It received no saved gate-position metadata. Gate-position agreement between encoder and replay was exact in the validation audit.

Each gate contained 120 RankCloak attempts and 120 paired length-matched ordinary controls across three model families, eight artifact classes, all 20 supported artifact/representation cells, and two templates. The ungated, moderate, and strict RankCloak conditions shared one stable seed per cell; controls used a separate shared ordinary-generation seed. Fixed-token-budget capacity was evaluated at the paired ungated span length for every attempt. Fixed-payload bits per generated token was defined only when the complete payload was embedded; completion rate and fixed-budget payload fraction are therefore separate endpoints.

Gate	Complete	Budget failures	Capacity, bit/token	Capacity n	Budget payload fraction	Length ratio	Saved-ID/visible recovery
Ungated	120/120	0	7.4001, conditional	120	1.0000	1.0000	120/120; 62/120
Moderate median	120/120	0	5.7920, conditional	120	0.7518	1.3743	120/120; 65/120
Strict 75th percentile	114/120	6	4.0513, conditional	114	0.4567	2.7615	114/120; 68/120

Table S23. Entropy-gate generation, capacity, length, and recovery. Capacity is fixed-payload bits per generated token conditional on completion. In particular, strict-gate capacity excludes six incomplete cases and has $n = 114$. Completion and recovery retain all 120 attempts in their denominators.

Strict completion by model was 40/40 for Llama, 39/40 for Mistral, and 35/40 for Qwen. The corresponding strict conditional capacity estimates were 6.1322 ($n = 40$), 2.9207 ($n = 39$), and 2.9331 ($n = 35$) bits/token. Visible-text recovery under ungated/moderate/strict conditions was 29/40, 29/40, and 34/40 for Llama; 0/40 in every Mistral condition; and 33/40, 36/40, and 34/40 for Qwen. The aggregate visible-text differences therefore do not establish a general gate benefit.

Relative to ungated output, moderate gating added a mean 20.1083 words, changed unique-word fraction by -0.0338 , and changed repeated-bigram fraction by $+0.0106$. Strict gating added 103.0083 words, changed unique-word fraction by -0.1431 , and changed repeated-bigram fraction by $+0.0423$. These are surface heuristics, not participant judgments. At forced payload positions, mean rank was 88.0552 for ungated and moderate gates and 89.4915 for the strict gate; mean forced-token rank pressure declined from 3.8585 to 3.4276 and 3.0308 nats. Gating changes the contexts in which payload ranks are consumed but does not change the payload-rank distribution for completed paired cells.

Entropy-gated RankCloak: capacity and recovery

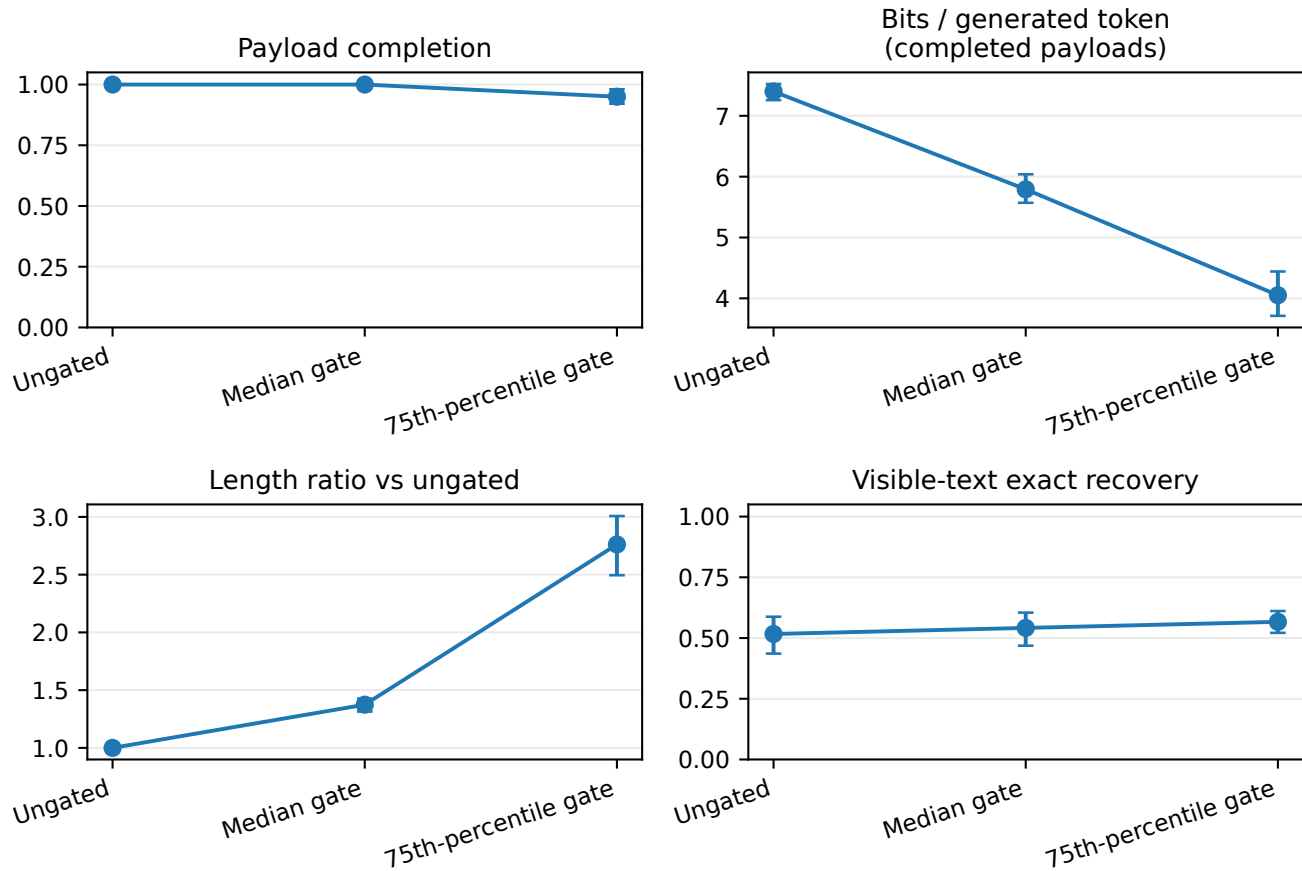


Figure S10. Entropy gating, capacity, length, completion, and recovery. Fixed-payload capacity is conditional on completion; fixed-budget payload fraction and completion retain all attempts.

340

The generated detector test was external to fitting. Historical train and validation rows carrying any of the 16 entropy-study payload instances were excluded, and exact/near-duplicate auditing was repeated before feature extraction. The audit retained 12,714/13,114 rows, removed 400, found zero cross-partition links, and left no condition unavailable. Gate-specific test controls numbered 117 ungated, 113 moderate, and 116 strict; none supports empirical 0.1% resolution.

Detector	Gate	Pos./neg.	ROC-AUC	0–1% pAUC	TPR/FPR at frozen 1% threshold	Exact FP
TextCNN	Ungated	117/117	0.9953	0.8536	0.8718/0.0085	1/117
TextCNN	Moderate	113/113	0.9825	0.7416	0.7965/0.0088	1/113
TextCNN	Strict	116/116	0.8850	0.4138	0.4310/0.0172	2/116
DeBERTa	Ungated	117/117	0.9999	0.9854	1.0000/0.0085	1/117
DeBERTa	Moderate	113/113	0.9785	0.6878	0.8496/0.0177	2/113
DeBERTa	Strict	116/116	0.8893	0.3484	0.4914/0.0259	3/116
Model-aware	Ungated	117/117	1.0000	1.0000	1.0000/0.0000	0/117
Model-aware	Moderate	113/113	0.9999	0.9922	0.9381/0.0000	0/113
Model-aware	Strict	116/116	0.9904	0.7845	0.6552/0.0000	0/116

Table S24. Detector behavior by entropy gate. Thresholds are the single historical-validation-frozen nominal 1% thresholds. Every gate has fewer than 1,000 test negatives, so empirical 0.1% TPR is unavailable.

Entropy-gate detector performance

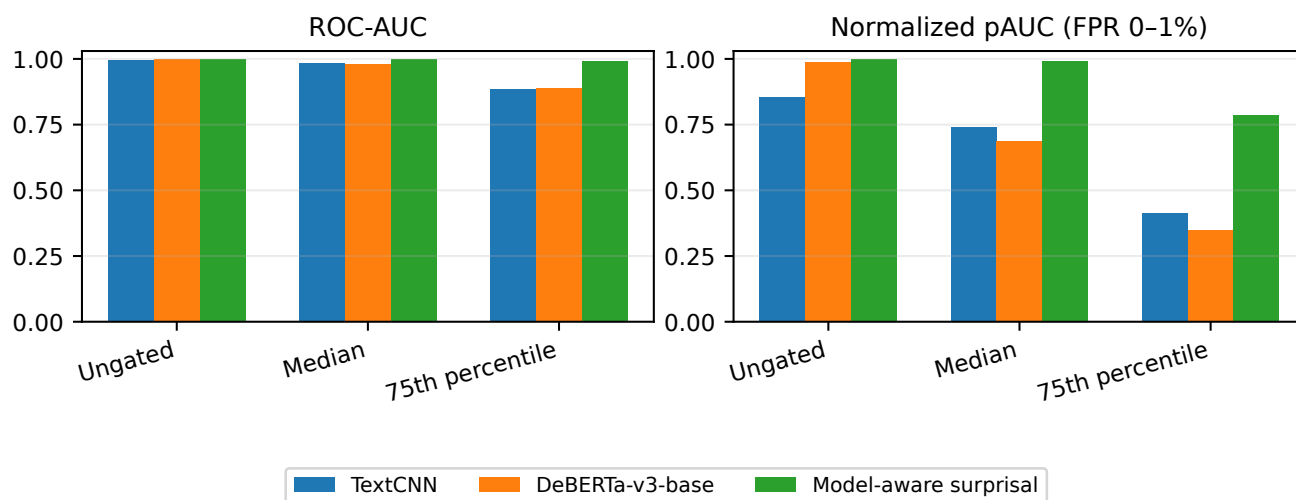


Figure S11. Entropy-gate detector performance. Stricter gating reduces text-only detectability in this bounded matrix but does not eliminate the strong exact-model-aware signal.

Taken together, stricter gating reduces text-only detection in the tested conditions while reducing capacity, substantially increasing length, producing six maximum-budget failures, and retaining exact-environment replay dependence. Model-aware ROC-AUC of 0.9904 under the strict gate is inconsistent with a claim of an undetectable channel.

Supplementary Note S14: Matched Q4 and Q8 quantization analysis

Pairing and recovery

The quantization analysis holds Qwen2.5-7B-Instruct revision 8911e8a47f92bac19d6f5c64a2e2095bd2f7d031, its embedded tokenizer, rendered prompt, payload, codec, safe-text filter, target length, temperature, top- p , llama-cpp-python 0.3.23 backend, and historical ordinary-control seed fixed. It successfully model-replayed all 1,920 historical Q4_K_M rows and completed all 1,920 paired Q8_0 generations. No additional Q4 covers were generated.

Visible-text recovery for the 960 historical Q4 RankCloak covers retokenized each recorded string under the exact embedded Q4 tokenizer. The 732 exact token-ID matches reused the validated saved-ID rank trace; the other 228 cases underwent rank replay through the pinned Q4 model with the saved embedding-span boundary. This remains replay-assisted and does not recover without the exact context and span contract. Q8 visible-text recovery used the corresponding exact model path. Pairing used frozen payload, representation, and prompt identities.

Recovery item	Q4_K_M	Q8_0	Paired interpretation
Saved-ID exact recovery	960/960	960/960	Perfect only under exact saved-ID replay
Visible-text exact recovery	732/960 (0.7625)	758/960 (0.7896)	Difference 0.0271; 95% CI [-0.0073, 0.0635]
Visible recovery 95% CI	[0.7344, 0.7896]	[0.7635, 0.8167]	Payload-group bootstrap, 2,000 resamples
Discordant outcomes	Q4 only 144	Q8 only 170	Exact two-sided McNemar $p = 0.1582$
Concordant outcomes	Both 588	Neither 58	No statistically supported Q8 advantage

Table S25. Paired Q4 and Q8 recovery. Rates use 960 frozen RankCloak pairs. The exact McNemar result is computed from the stored 144/170 discordant counts.

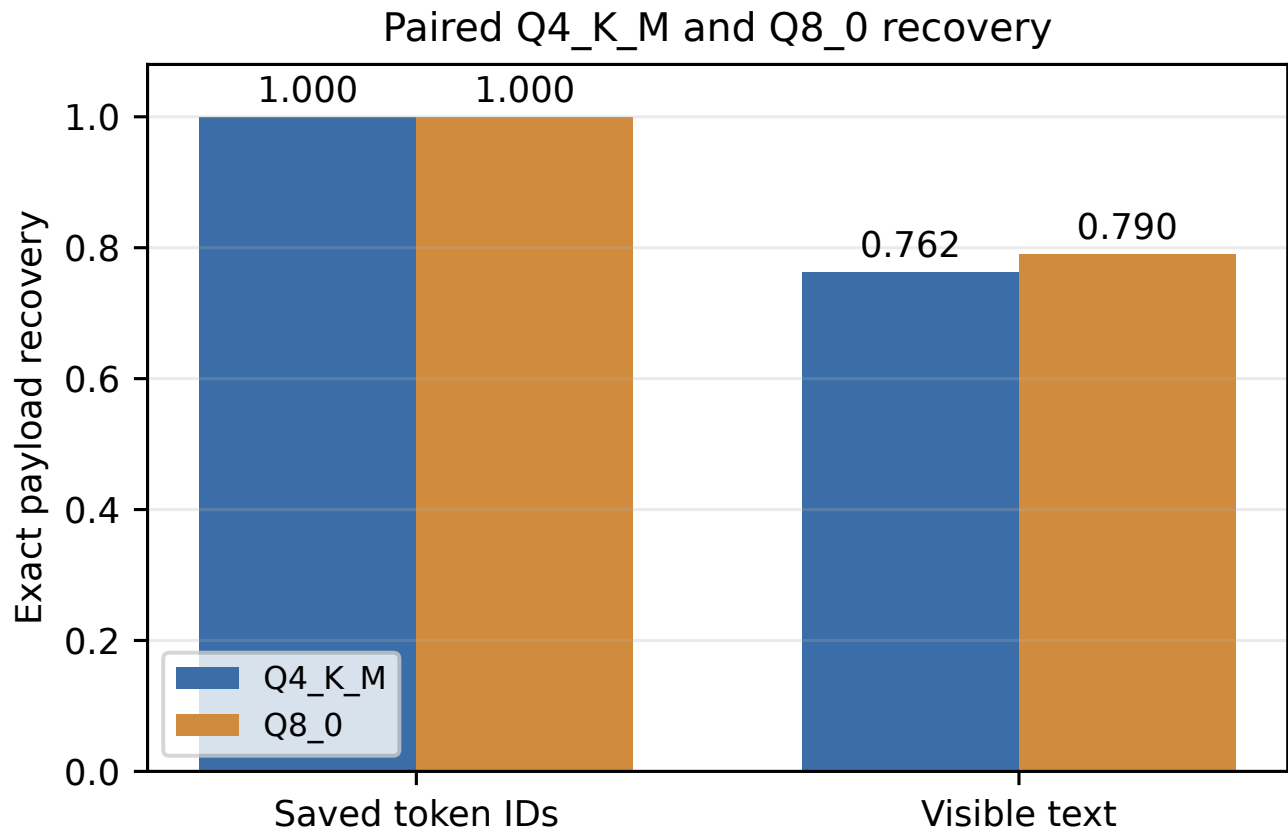


Figure S12. Paired saved-ID and visible-text recovery by quantization. The grouped interval for the paired difference includes zero.

On the identical historical Q4 RankCloak token path, Q8 minus Q4 mean next-token entropy was -0.0154 bits (95% CI $[-0.0187, -0.0120]$); the observed-token rank changed at a mean fraction 0.4639 of positions and the greedy token changed at 0.0884 . Independently generated Q4/Q8 RankCloak paths never matched exactly and had mean positionwise token match 0.0155 . These differences show substantial path sensitivity; they are not evidence that either quantization is generally more recoverable.

Matched Q4_K_M versus Q8_0 sensitivity

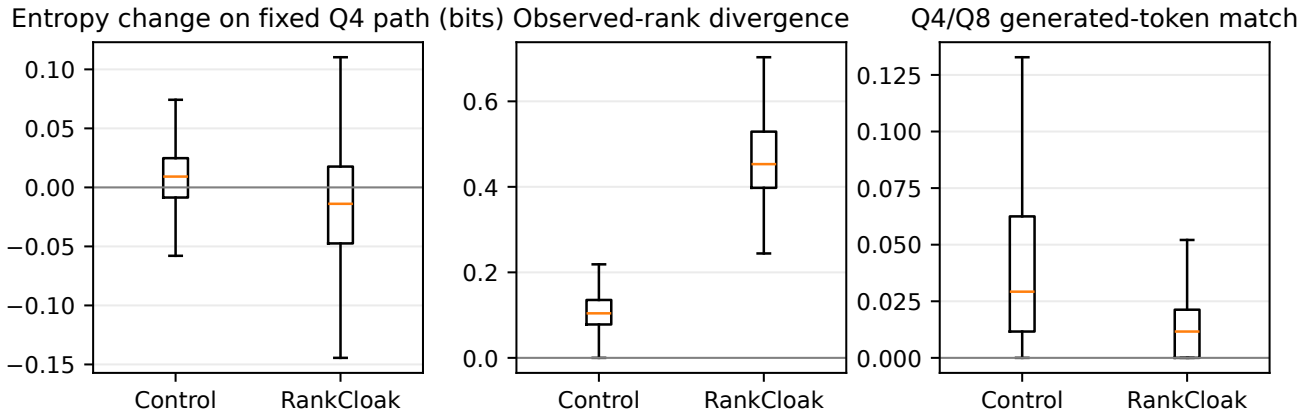


Figure S13. Matched quantization sensitivity. Same-path likelihood/rank comparisons and independently generated token-path agreement are paired within one pinned Qwen revision.

Cross-quantization detector transfer

Quantization detectors used the frozen payload train/validation/test assignment. Q4-to-Q8 fit and selected thresholds on Q4 before testing Q8; Q8-to-Q4 reversed the direction; the pooled fit used both quantizations in training and validation and tested only held payloads. Exact and declared near-duplicate audits preceded feature extraction and retained every row: 1,920/1,920 in each directional corpus and 3,840/3,840 pooled, with zero audited leakage.

Detector	Transfer	ROC-AUC	0–1% pAUC	TPR/FPR at frozen 1% threshold	Exact FP
TextCNN	Q4 to Q8	0.9927	0.7539	0.9375/0.0365	7/192
DeBERTa	Q4 to Q8	1.0000	1.0000	1.0000/0.0000	0/192
Model-aware	Q4 to Q8	1.0000	1.0000	1.0000/0.0000	0/192
TextCNN	Q8 to Q4	0.9944	0.8298	0.9479/0.0260	5/192
DeBERTa	Q8 to Q4	1.0000	1.0000	1.0000/0.0156	3/192
Model-aware	Q8 to Q4	1.0000	1.0000	0.9792/0.0000	0/192
TextCNN	Pooled	0.9965	0.8257	0.9609/0.0234	9/384
DeBERTa	Pooled	1.0000	1.0000	1.0000/0.0078	3/384
Model-aware	Pooled	1.0000	1.0000	0.9948/0.0000	0/384

Table S26. Cross-quantization and pooled detector behavior. Frozen nominal thresholds can drift in test. No test has 1,000 negatives, so empirical 0.1% results are unavailable.

Matched-quantization detector transfer

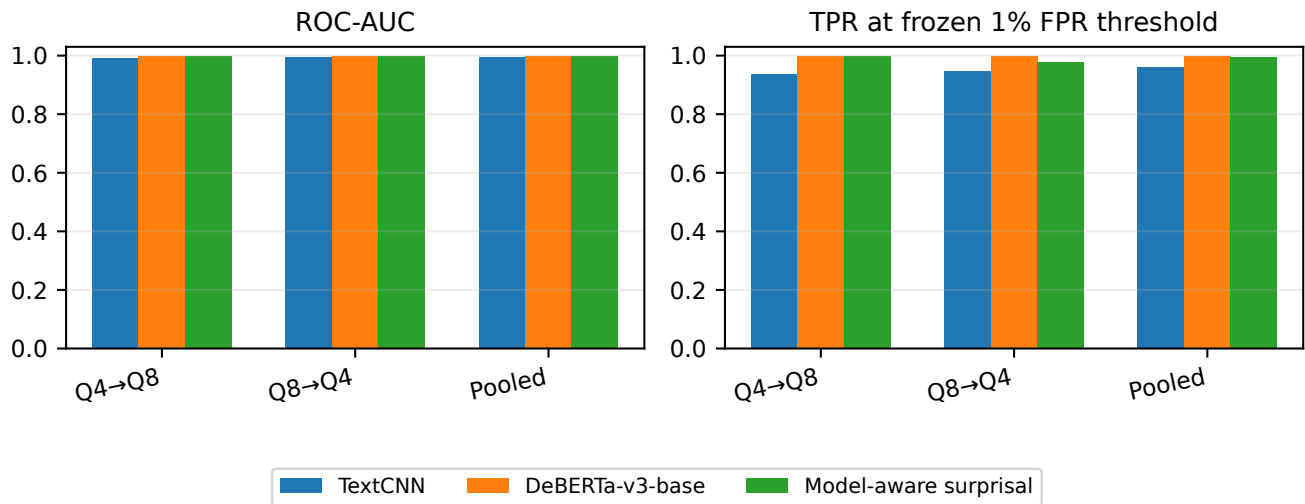


Figure S14. Detector transfer across Q4_K_M and Q8_0. Detection remains strong in both directions, while observed FPR can exceed the validation target.

The analysis isolates two quantizations of one model revision under one inference backend. It does not generalize to other Qwen revisions, architectures, bit widths, quantizers, backends, or hardware. In particular, the data do not justify stating that Q8 is generally more recoverable than Q4.

Supplementary Note S15: V3 provenance and remaining boundaries

The V3 run manifest binds the detector inputs, model-backed generations, artifact hashes, backend, model and tokenizer identities, partitions, and validation status. The final validation report records `status:pass`, exact artifact-manifest path and checksum validation for 5,833 entries, 24 fit-metadata files, 24 prediction files, strict leakage-audit success, conditional entropy-capacity validation, and paired quantization-recovery source validation. The generation-analysis validation records 18/18 calibration traces, 720/720 entropy generations, 1,920/1,920 Q4 replays, 960/960 Q4 visible-recovery records, 1,920/1,920 Q8 generations, and zero failure records. Machine-readable tables and full provenance are retained under `results/revision_v3`; the manuscript reports from those sealed outputs rather than rerunning generation.

Table S27. Remaining boundaries after the targeted V3 evaluations.

Issue	Added evidence	Boundary that remains
Saved-token condition	Perfect primary and paired-quantization recovery from the preserved token stream under recorded configurations	Tests token-preserving delivery or experimental retained IDs under the predeclared protocol; distinct from ordinary visible-text communication
Visible-text transport	88/144 historical robustness recovery; 732/960 Q4 and 758/960 Q8 in one paired matrix	Paraphrase, markdown/format transfer, cross-model decoding, and retokenization remain fragile or unsuccessful
Strict deduplication	Component-safe split with zero declared cross-partition links	Does not detect semantic paraphrases below the 0.95 character- <i>n</i> -gram threshold
Human-authored controls	250 frozen Dolly negatives across 16 templates	Not a participant study; sparse genre coverage and generic machine/human signal remain
Unseen families	Three leave-one-family-out tests	Not every architecture, training corpus, model size, proprietary system, or decoding stack

Issue	Added evidence	Boundary that remains
Adaptive detection	Two data-adaptive text-only detectors and an exact-model-aware likelihood attacker	Strong but not exhaustive attackers or deployment distributions
Low-FPR reporting	Exact counts and validation-frozen thresholds; matched 0.1% estimates where supported	Human, family, entropy, and quantization tests have fewer than 1,000 negatives and no empirical 0.1% resolution
Entropy gating	120 attempts per gate; capacity, length, recovery, quality, and detector outcomes	Two templates; thresholds from 768 positions/model; strict capacity conditional on 114 completions; exact-model AUC remains 0.9904
Quantization	Paired Q4_K_M/Q8_0 recovery and bidirectional detector transfer	One pinned Qwen revision, two quantizations, one backend; no supported general Q8 advantage
Naturalness	Automated surface/model diagnostics and a human-authored control corpus	No participants or human ratings; controls are not evidence of human-perceived naturalness
Deployment/security	Failure surfaces and strong adverse detectability are documented	No real-world deployment, robust channel, secrecy proof, cryptographic protection, or undetectability

380 The bounded evaluations directly address the requested controls, leakage prevention, held-family transfer, adaptive and model-aware attackers, low-FPR reporting, entropy gating, and quantization sensitivity. They refine the evidence for an artifact-specific hiding and recovery method: the artifact surface form is concealed, exact recovery is configuration-dependent, visible-text transport is weaker, and current steganalysis often detects the encoding.

References

- 385 1. Cai, S., Ding, L. & Tao, D. Entropy-guided watermarking for LLMs: A test-time framework for robust and traceable text generation. *arXiv preprint arXiv:2504.12108* DOI: [10.48550/arXiv.2504.12108](https://doi.org/10.48550/arXiv.2504.12108) (2025).